



US012399589B2

(12) **United States Patent**
Chen et al.

(10) **Patent No.:** **US 12,399,589 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **DISPLAY SUBSTRATE AND DISPLAY DEVICE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **17/800,045**

(22) PCT Filed: **Nov. 4, 2021**

(86) PCT No.: **PCT/CN2021/128698**

§ 371 (c)(1),

(2) Date: **Aug. 16, 2022**

(87) PCT Pub. No.: **WO2022/242048**

PCT Pub. Date: **Nov. 24, 2022**

(65) **Prior Publication Data**

US 2024/0179989 A1 May 30, 2024

(30) **Foreign Application Priority Data**

May 19, 2021 (WO) PCT/CN2021/094676

Jun. 29, 2021 (CN) 202110726472.5

(51) **Int. Cl.**
H10K 59/35 (2023.01)
G06F 3/041 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **G06F 3/04164** (2019.05); **G06F 3/0445** (2019.05); **G06F 3/047** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC H10K 59/121; H10K 59/1213
See application file for complete search history.

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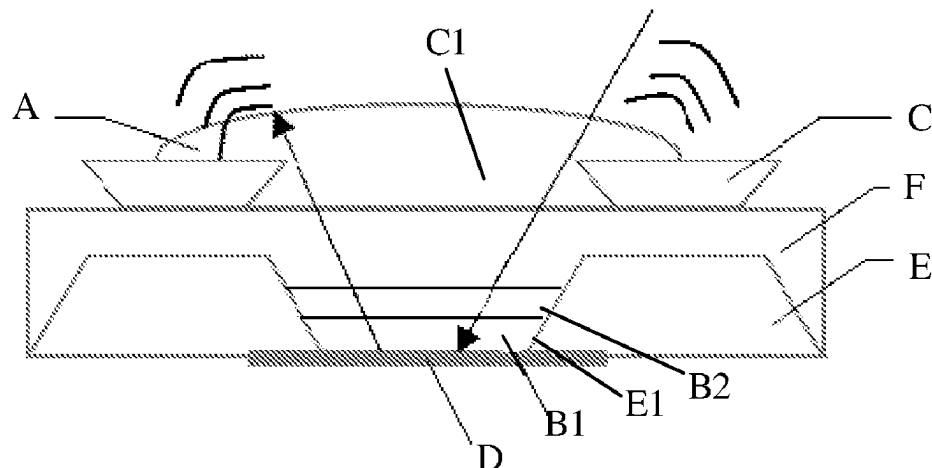
Primary Examiner — Evren Seven

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William Collard

(57) **ABSTRACT**

A display substrate and a display device are disclosed, and the display substrate has a plurality of sub-pixels, and includes a base substrate, a driving circuit layer, a pixel definition layer and a light-emitting device layer, and a black matrix layer. Each sub-pixel includes a pixel driving circuit in the driving circuit layer and a light-emitting device in the light-emitting device layer. The black matrix layer includes a plurality of first light-transmitting openings respectively

(Continued)



exposing the light-emitting devices of the plurality of sub-pixels in a direction perpendicular to a surface of the base substrate. The pixel definition layer includes a plurality of sub-pixel openings, at least part of the plurality of sub-pixel openings are in one-to-one correspondence with and at least partially overlapped with the plurality of first light-transmitting openings, and at least one of the plurality of first light-transmitting openings has an arc-shaped edge.

19 Claims, 17 Drawing Sheets

(51) Int. Cl.

G06F 3/044 (2006.01)
G06F 3/047 (2006.01)
H10K 59/122 (2023.01)
H10K 59/131 (2023.01)
H10K 59/65 (2023.01)
H10K 59/80 (2023.01)

(52) U.S. Cl.

CPC **H10K 59/122** (2023.02); **H10K 59/352** (2023.02); **H10K 59/353** (2023.02); **H10K 59/873** (2023.02); **H10K 59/8792** (2023.02); **H10K 59/131** (2023.02); **H10K 59/65** (2023.02)

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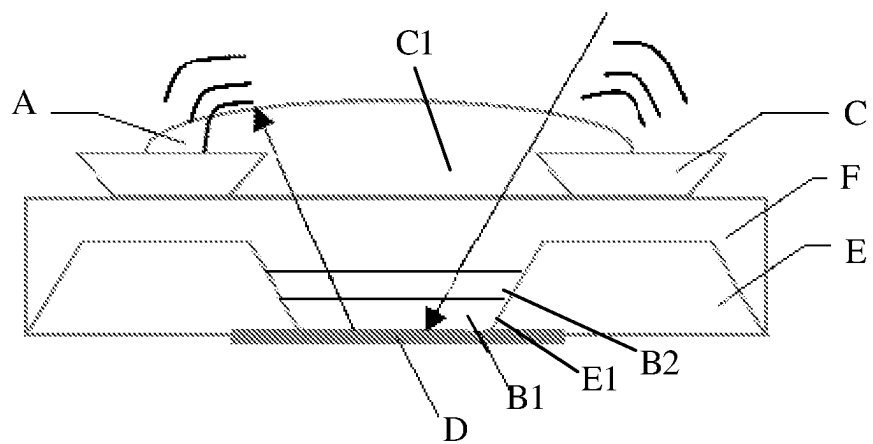


FIG. 1

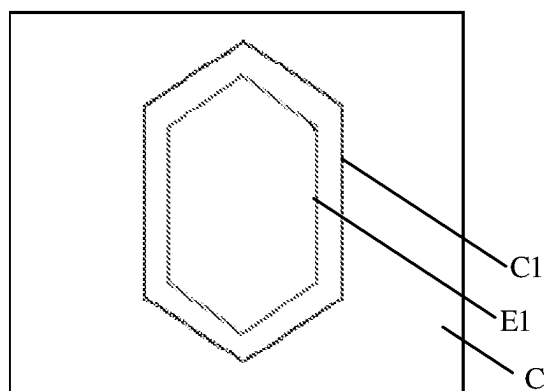


FIG. 2

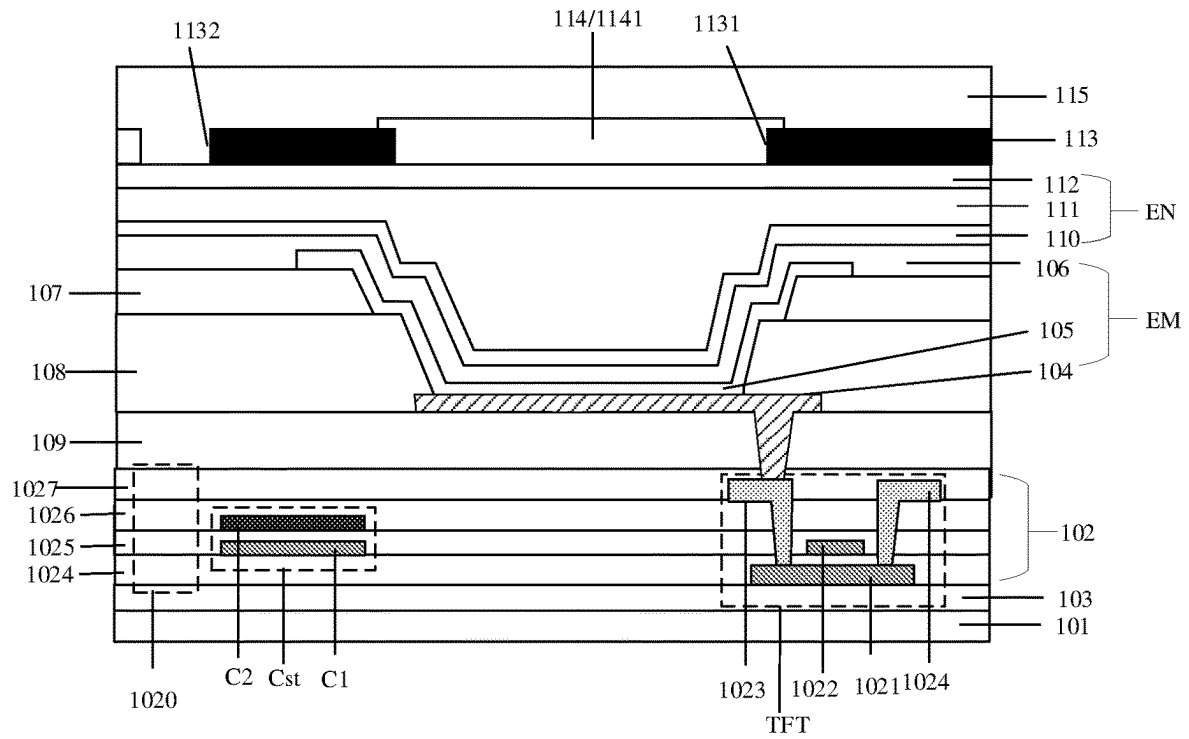


FIG. 3

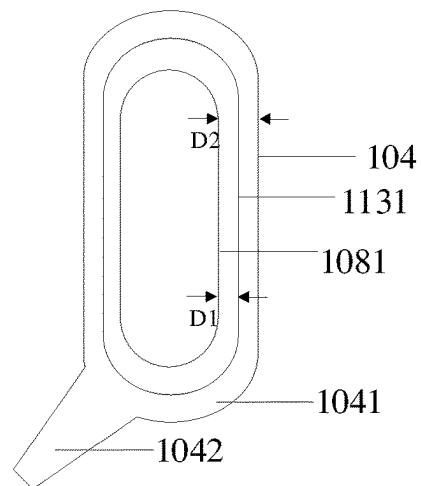


FIG. 4A

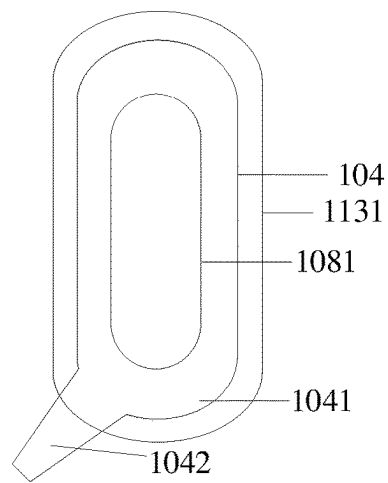


FIG. 4B

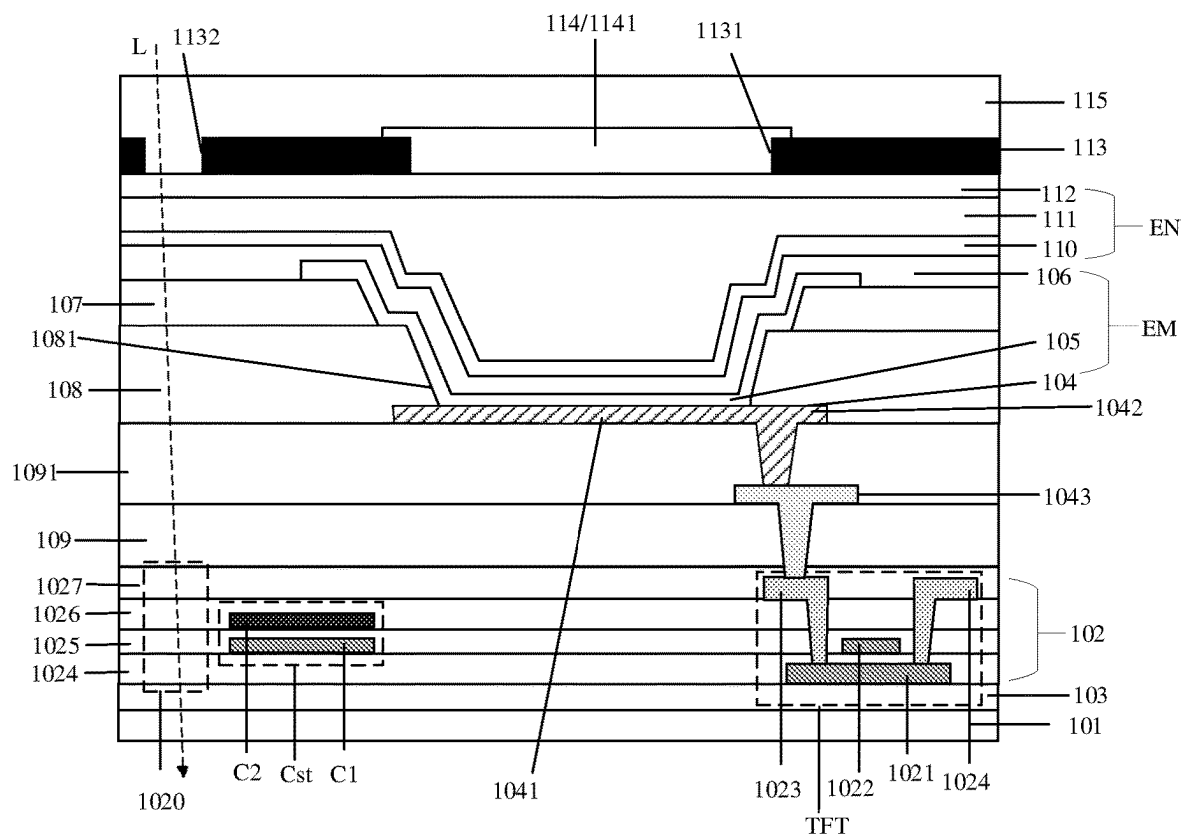


FIG. 5

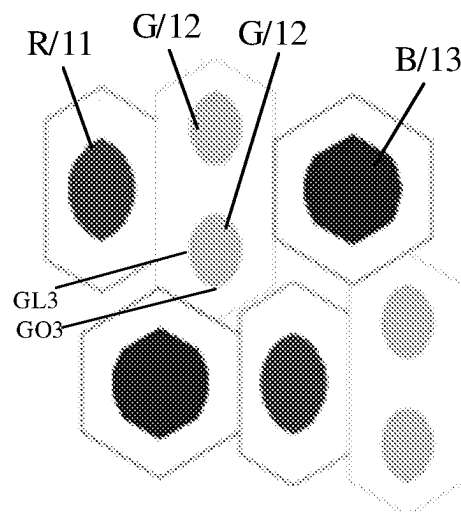


FIG. 6

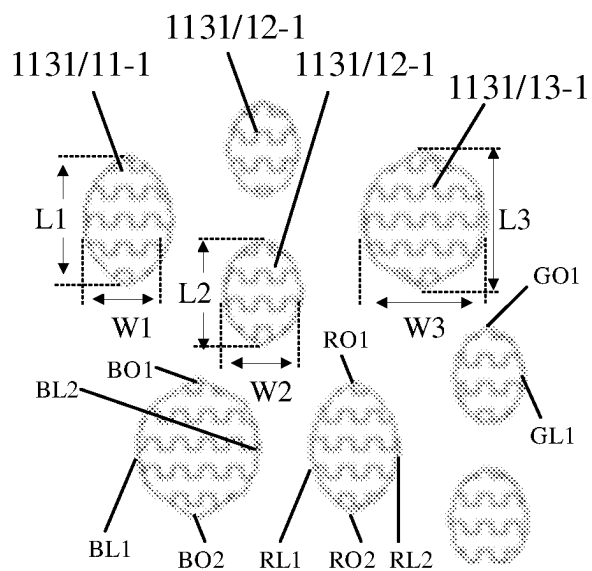


FIG. 7

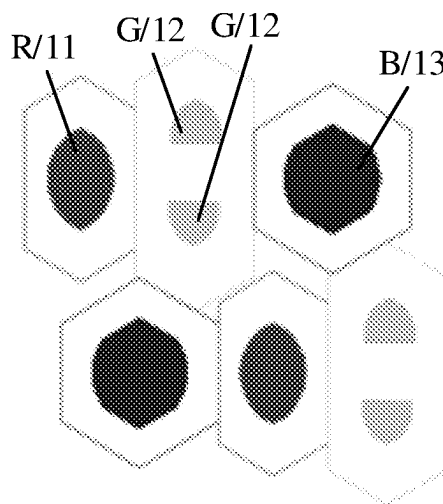


FIG. 8A

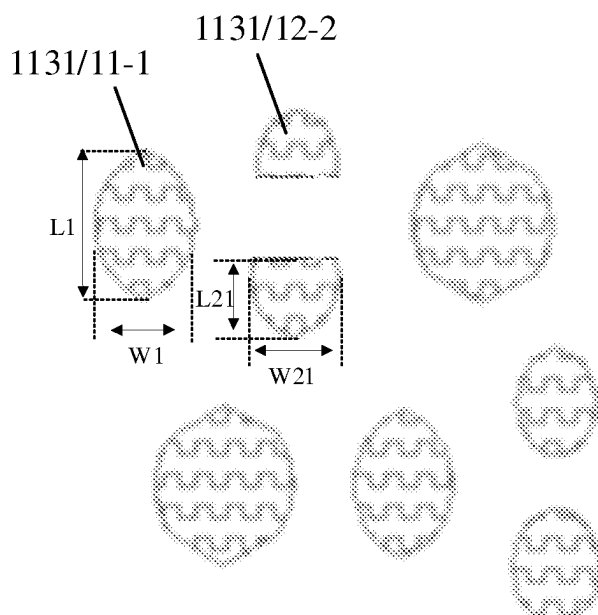


FIG. 8B

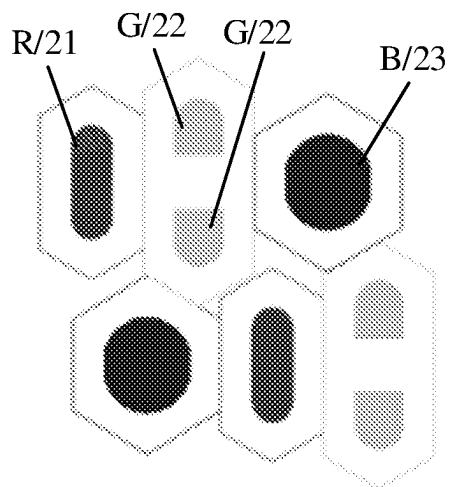


FIG. 9

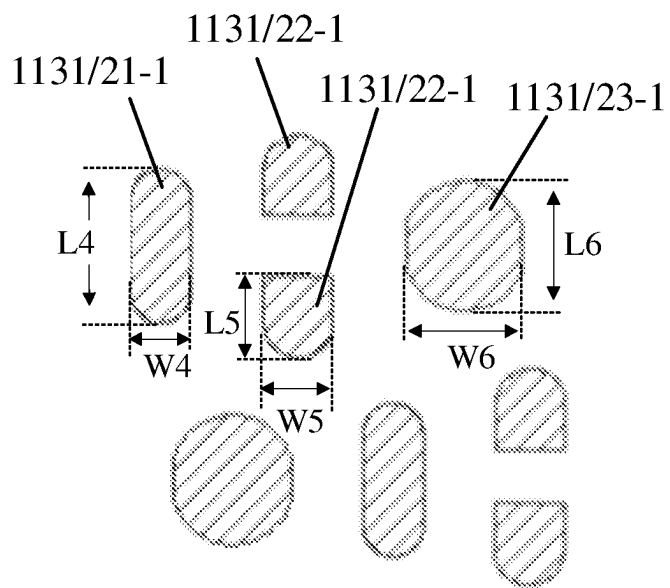


FIG. 10

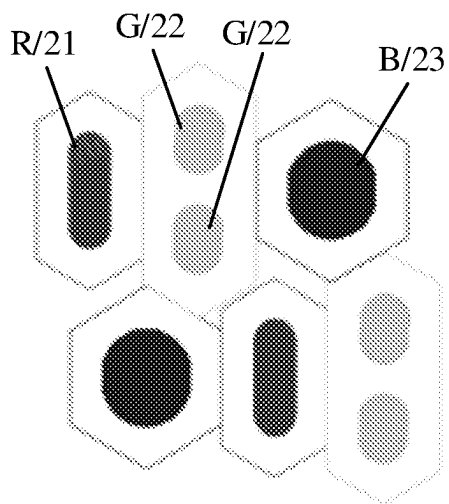


FIG. 11A

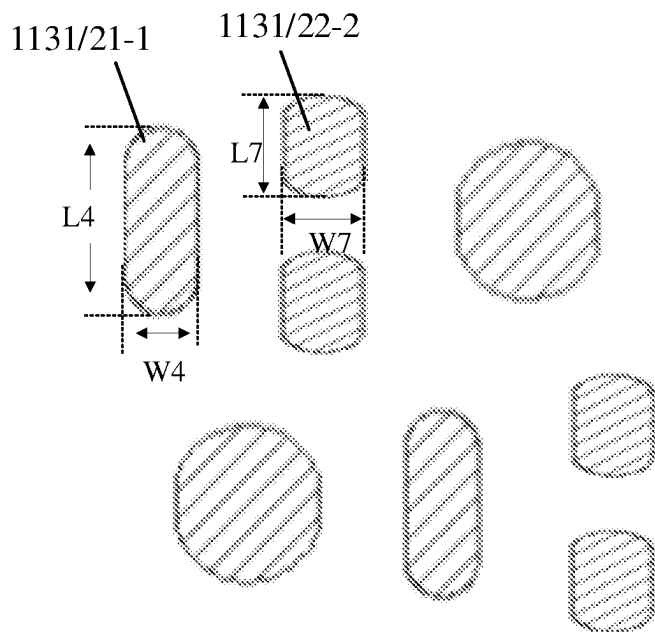


FIG. 11B

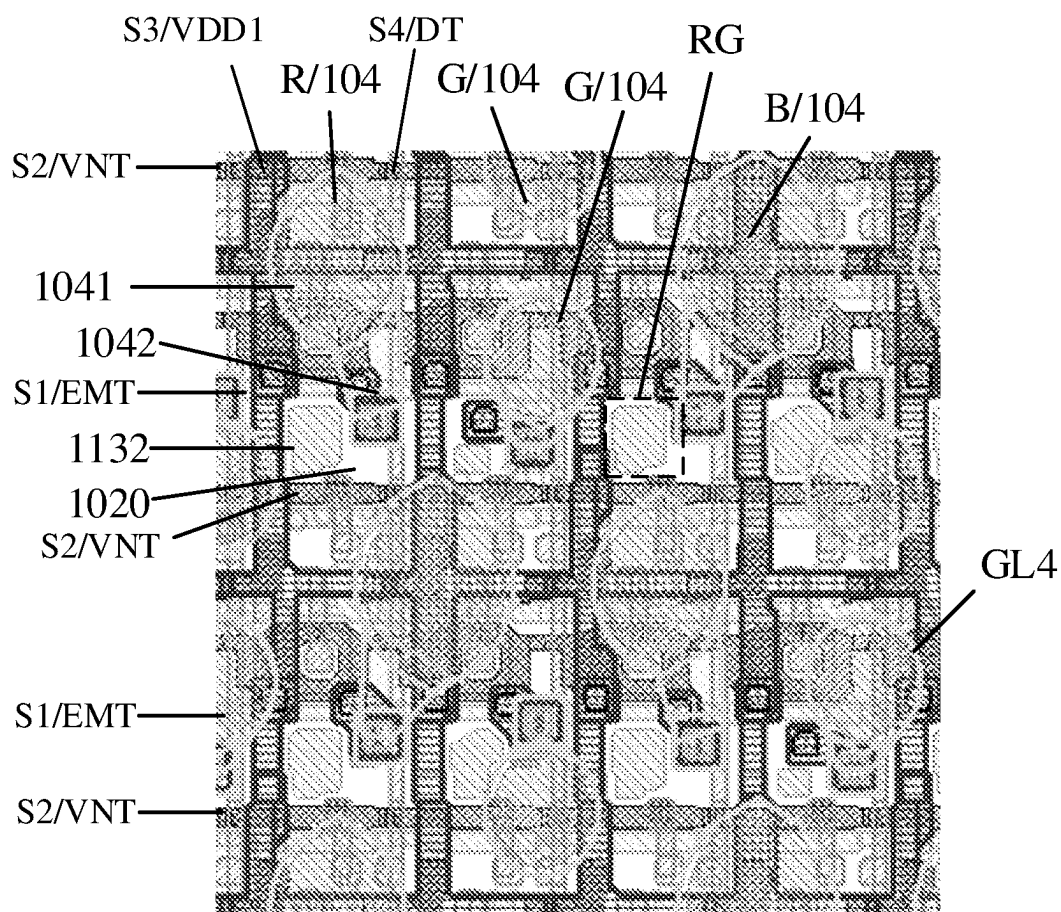


FIG. 12

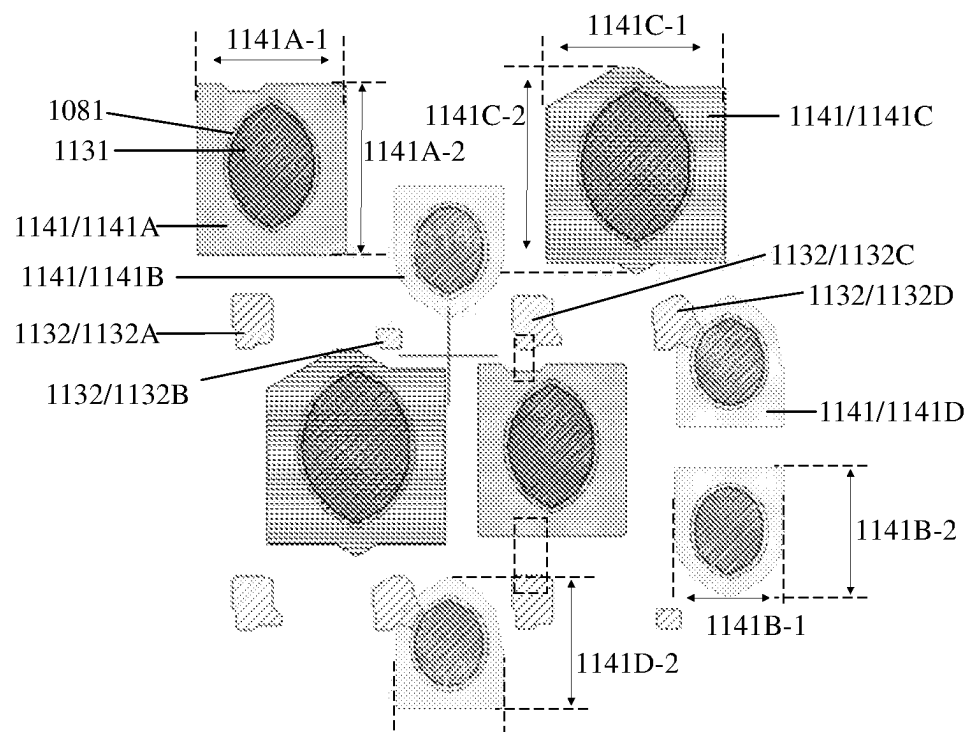


FIG. 13

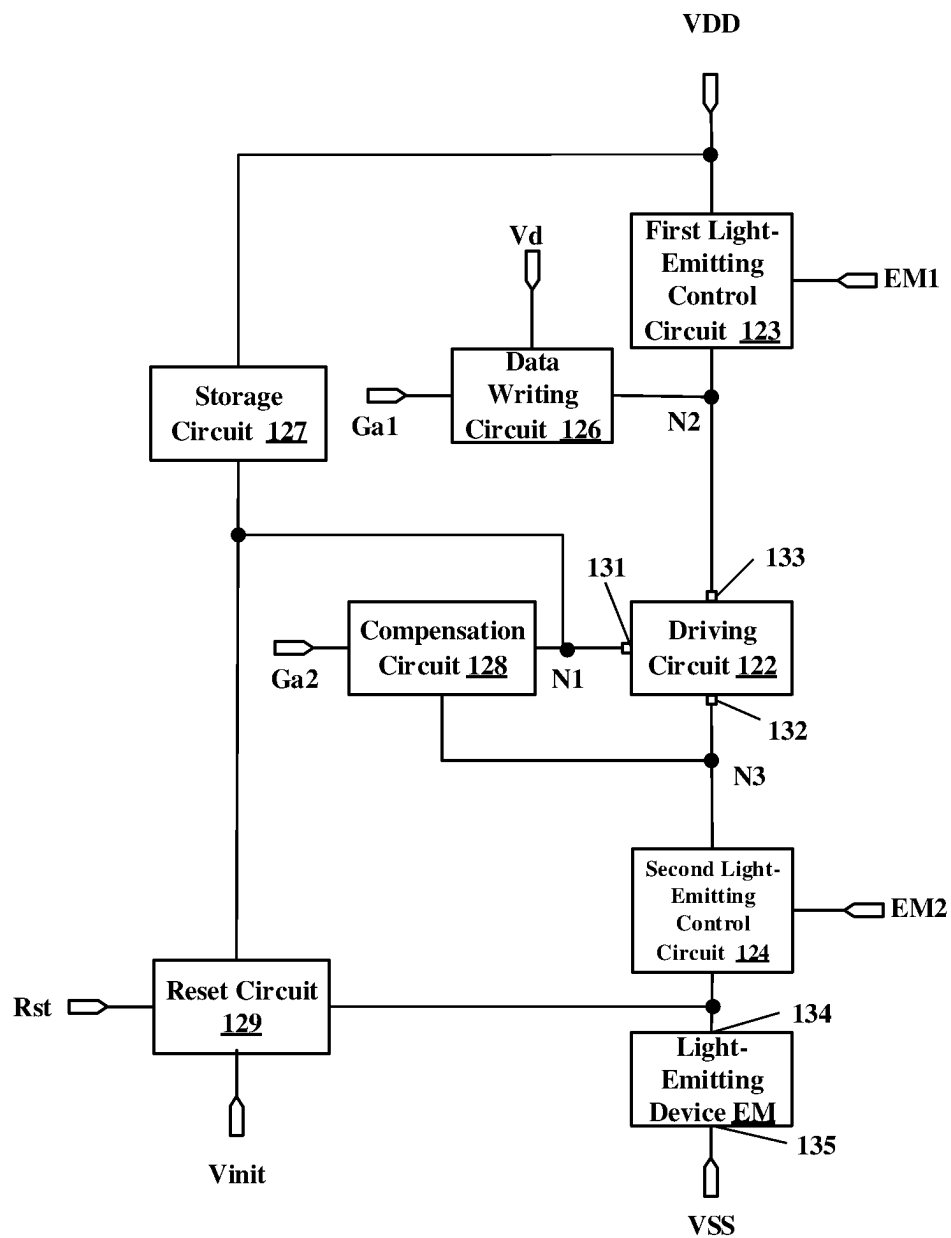


FIG. 14A

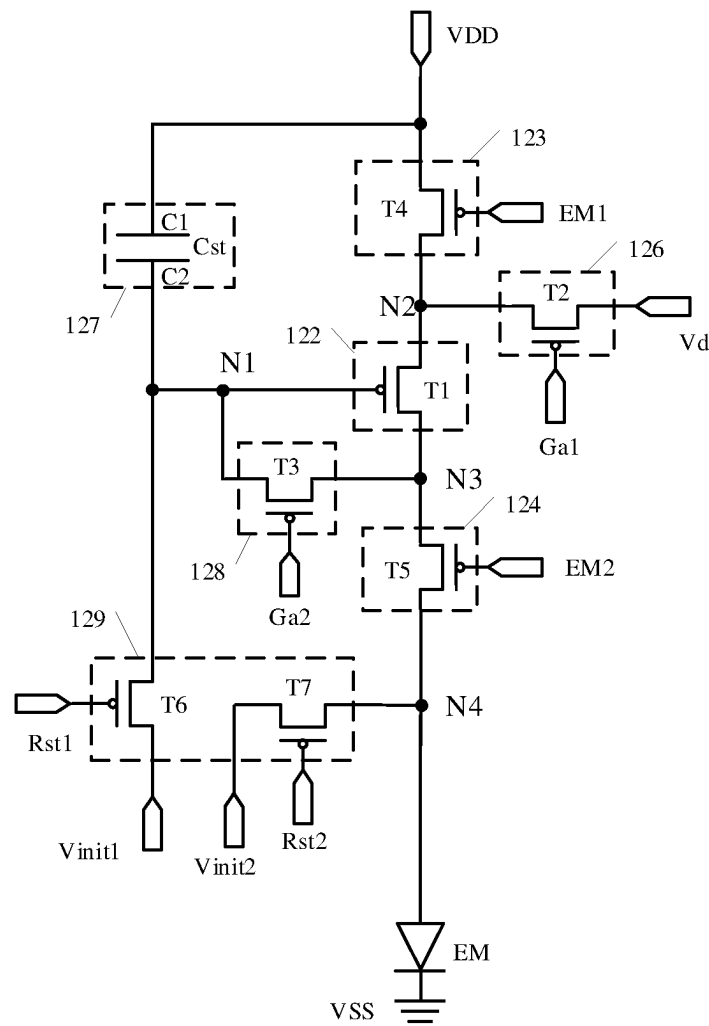


FIG. 14B

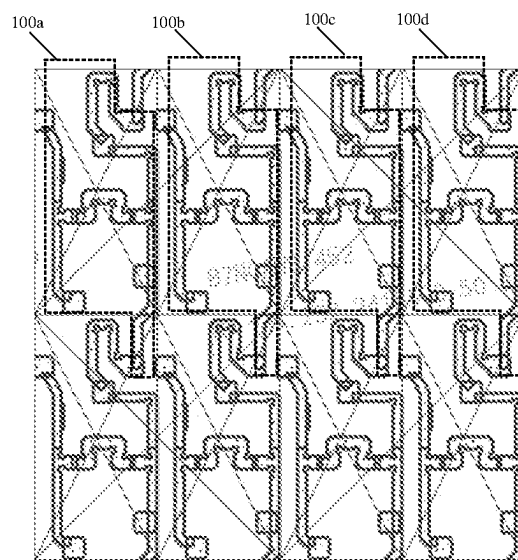


FIG. 15

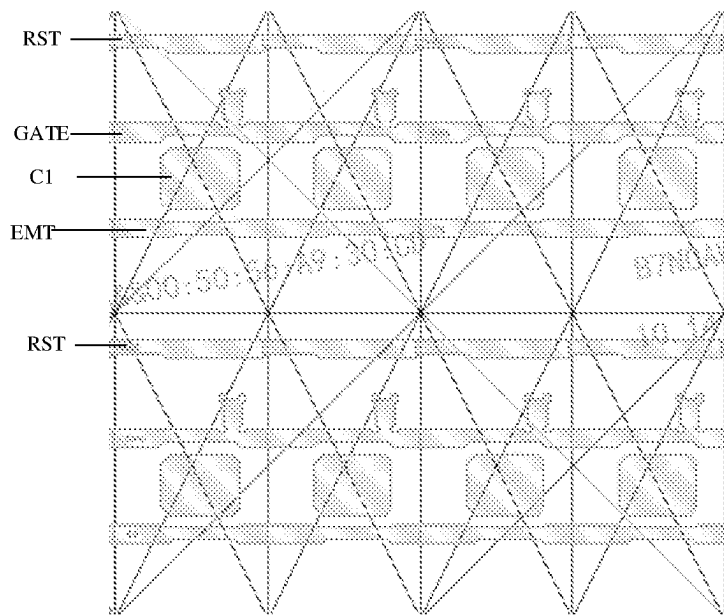


FIG. 16A

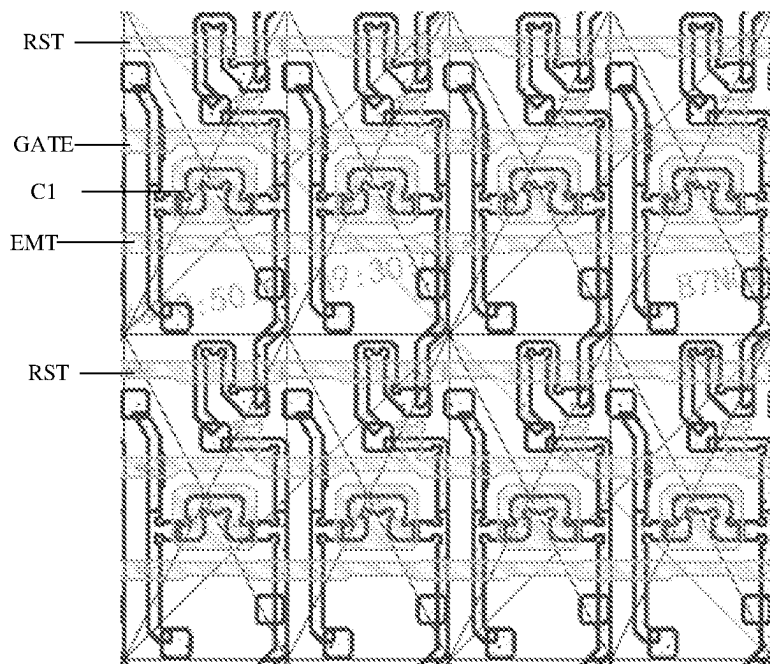


FIG. 16B

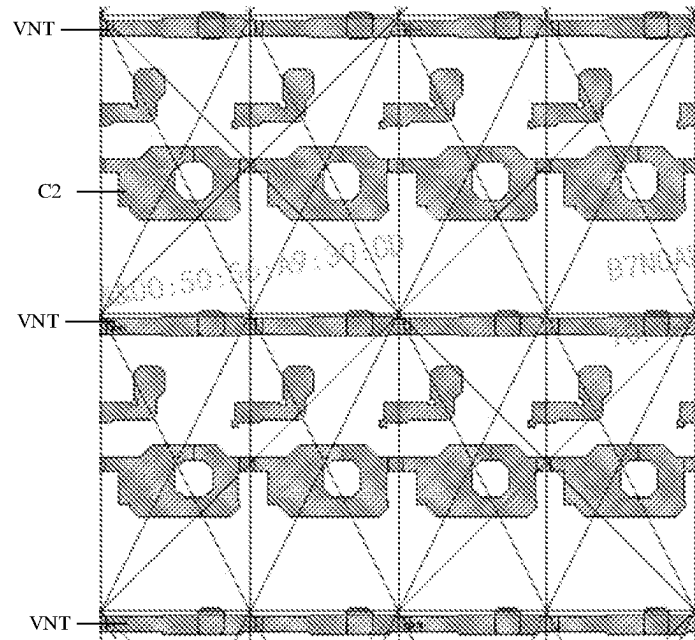


FIG. 17A

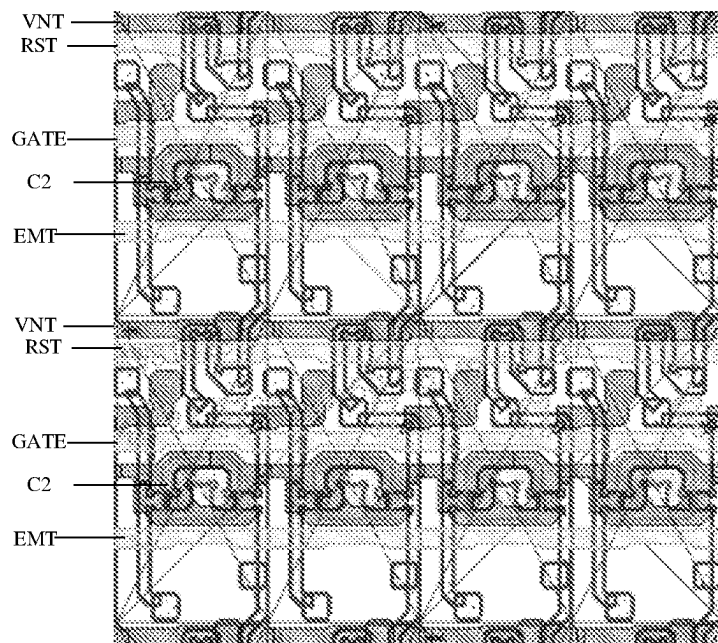


FIG. 17B

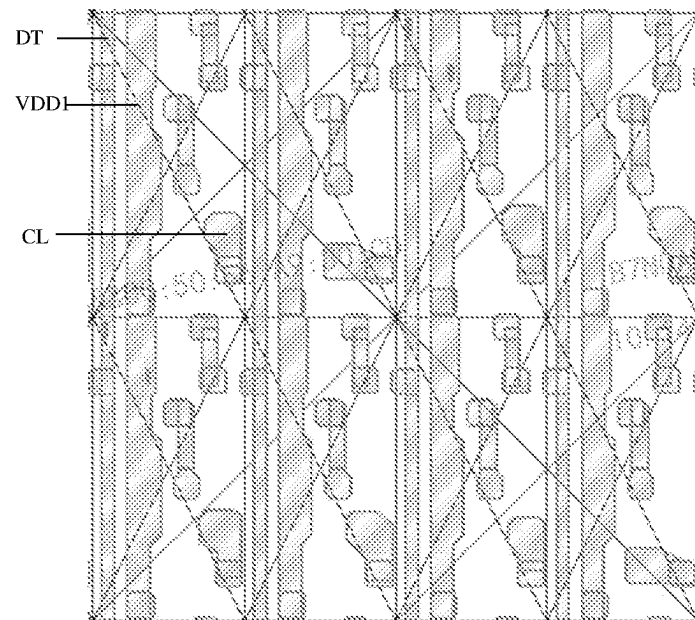


FIG. 18A

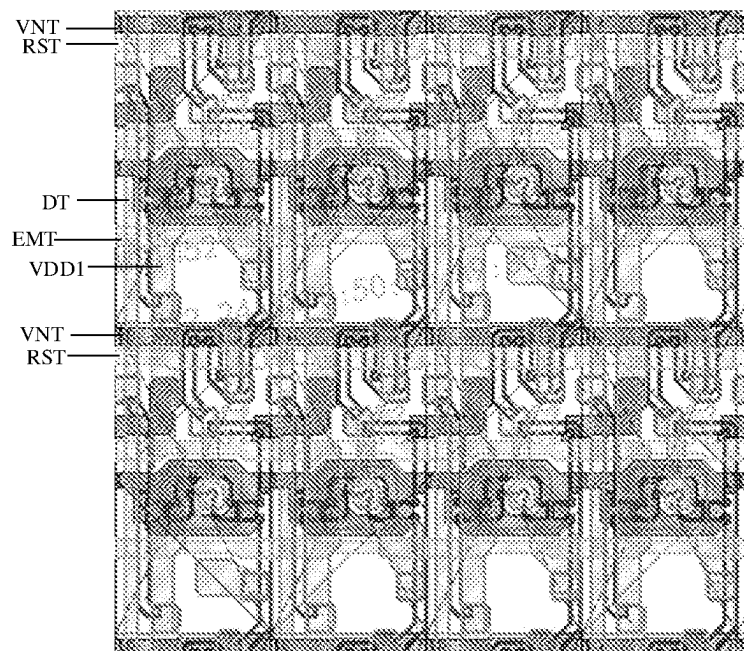


FIG. 18B

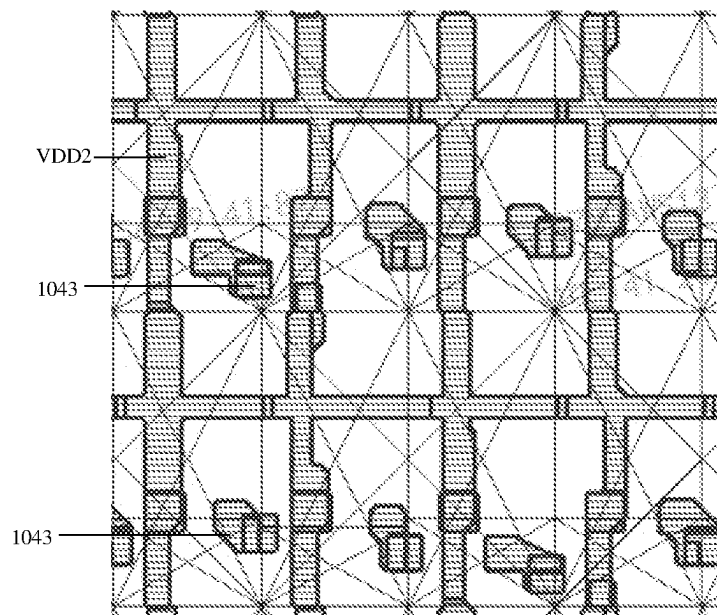


FIG. 19A

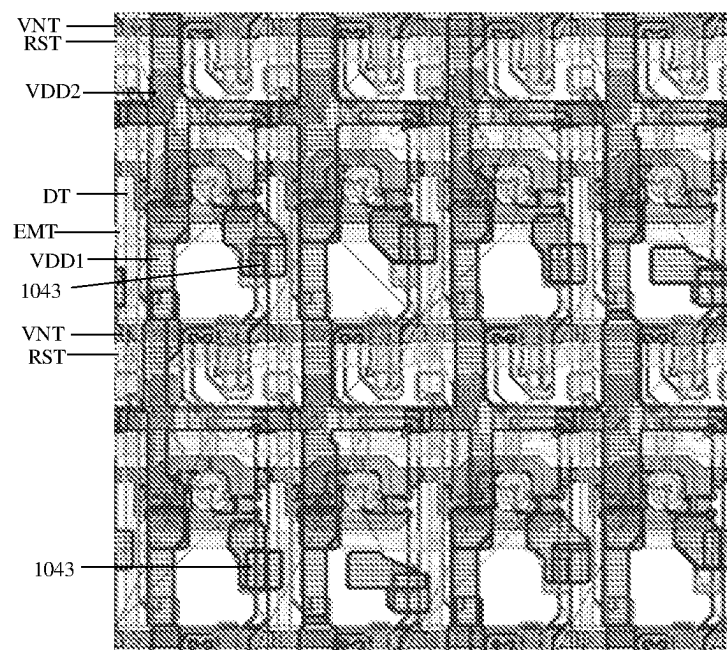


FIG. 19B

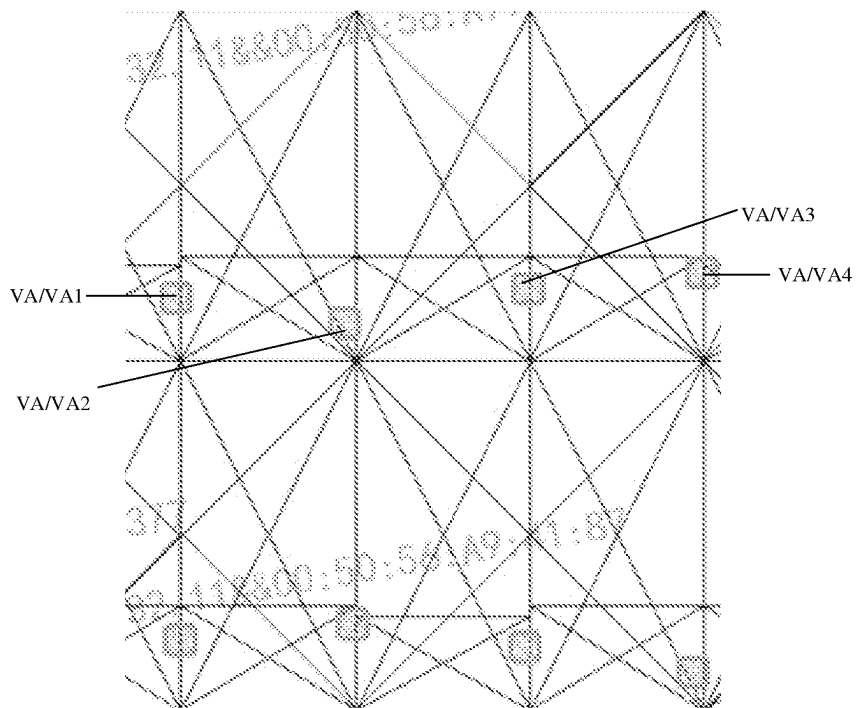


FIG. 20

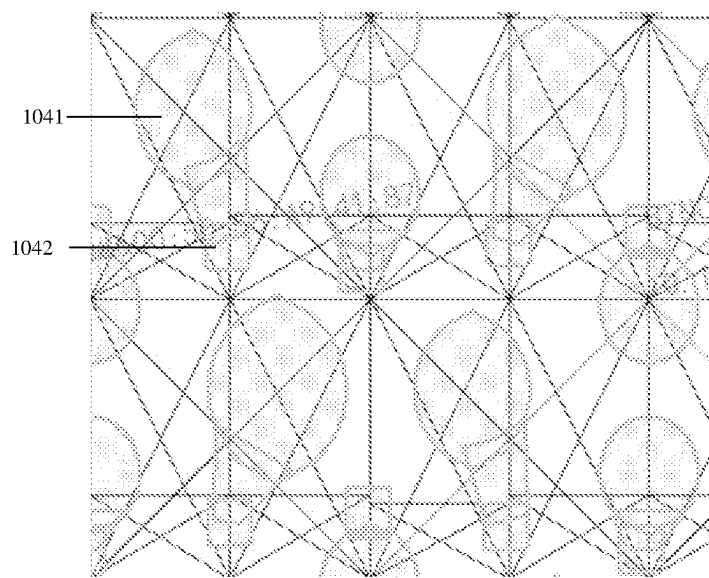


FIG. 21A

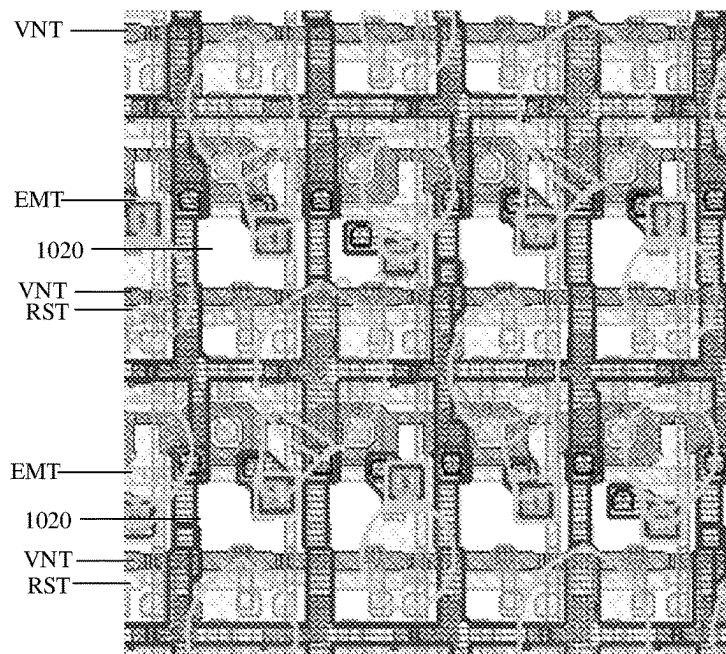


FIG. 21B

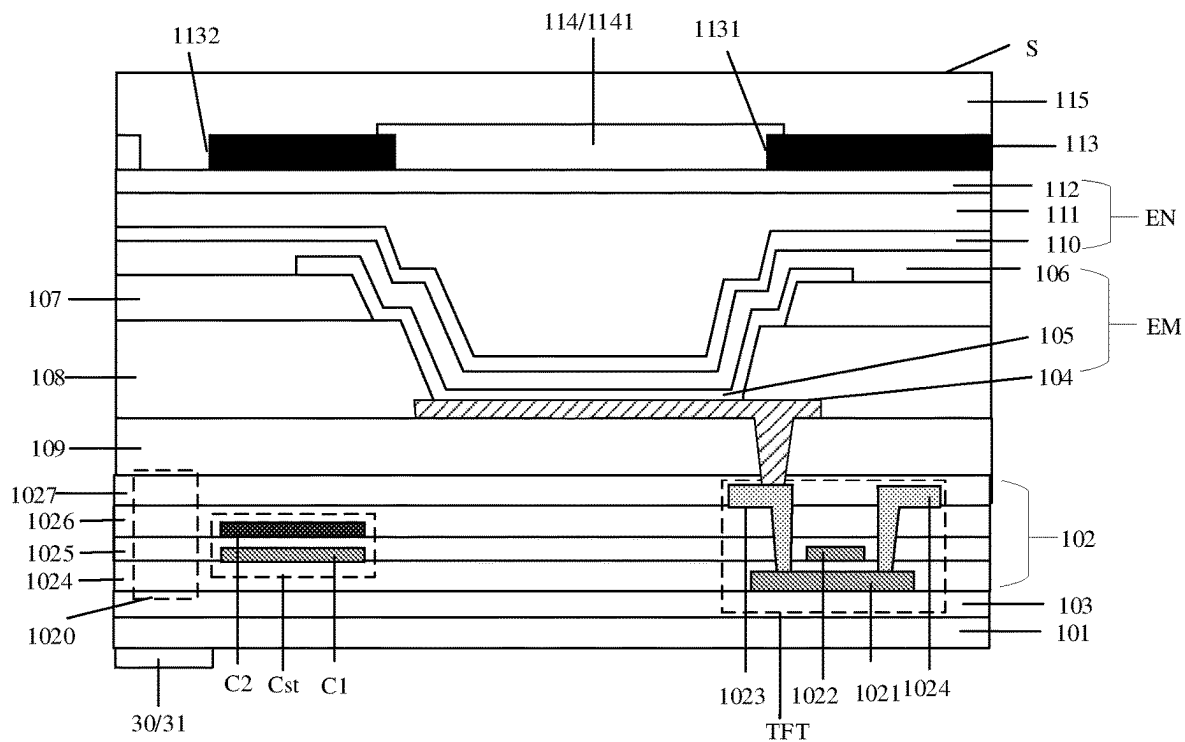


FIG. 22

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**DISPLAY SUBSTRATE AND DISPLAY
DEVICE****CROSS REFERENCE TO RELATED
APPLICATIONS**

This application is the National Stage of PCT/CN2021/128698 filed on Nov. 4, 2021, which claims priority under 35 U.S.C. § 119 to International Patent Application No. PCT/CN2021/094676, filed on May 19, 2021, and Chinese Patent Application No. 202110726472.5, filed on Jun. 29, 2021, the entire disclosure of which is incorporated herein by reference as part of the present application.

TECHNICAL FIELD

Embodiments of the present disclosure relate to a display substrate and a display device.

BACKGROUND

OLED (Organic Light Emitting Diode) display device has a series of advantages such as self-luminescence, high contrast, high definition, wide viewing angle, low power consumption, fast response speed, and low manufacturing cost, and has become one of the key development directions of the new generation of display devices, so the OLED display device has received more and more attention.

SUMMARY

At least one embodiment of the present disclosure provides a display substrate, the display substrate has a plurality of sub-pixels arranged in an array, and comprises a base substrate, a driving circuit layer on the base substrate, a pixel definition layer and a light-emitting device layer on a side of the driving circuit layer away from the base substrate, and a black matrix layer on a side of the light-emitting device layer away from the base substrate, each of the plurality of sub-pixels comprises a pixel driving circuit in the driving circuit layer and a light-emitting device in the light-emitting device layer, and the pixel driving circuit is configured to drive the light-emitting device; the pixel definition layer comprises a plurality of sub-pixel openings, the light-emitting device comprises a first electrode layer, a light-emitting material layer, and a second electrode layer that are sequentially stacked in a direction away from the base substrate, the pixel definition layer is on a side of the first electrode layer away from the base substrate, and the plurality of sub-pixel openings respectively expose first electrode layers of light-emitting devices of the plurality of sub-pixels; the black matrix layer includes a plurality of first light-transmitting openings respectively exposing the light-emitting devices of the plurality of sub-pixels in a direction perpendicular to a surface of the base substrate, and at least one of the plurality of first light-transmitting openings has an arc-shaped edge; and in the direction perpendicular to the surface of the base substrate, at least part of the plurality of sub-pixel openings are in one-to-one correspondence with and at least partially overlapped with the plurality of first light-transmitting openings.

For example, in the display substrate provided by at least one embodiment of the present disclosure, in a direction parallel to the surface of the base substrate, a plane shape of at least one of the plurality of first light-transmitting openings is elliptical, semi-elliptical, circular, semi-circular, race-track-shaped, or semi-racetrack-shaped.

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For example, in the display substrate provided by at least one embodiment of the present disclosure, in a direction parallel to the surface of the base substrate, a plane shape of at least one of the plurality of sub-pixel openings is elliptical, semi-elliptical, circular, semi-circular, racetrack-shaped, or semi-racetrack-shaped.

For example, in the display substrate provided by at least one embodiment of the present disclosure, for a sub-pixel opening and a first light-transmitting opening that are provided corresponding to each other, and in a direction parallel to the surface of the base substrate, a plane shape of the sub-pixel opening is same as a plane shape of the first light-transmitting opening.

For example, in the display substrate provided by at least one embodiment of the present disclosure, an orthographic projection of the sub-pixel opening on the base substrate is located within an orthographic projection of the first light-transmitting opening on the base substrate.

For example, in the display substrate provided by at least one embodiment of the present disclosure, a minimum distance between an edge of the orthographic projection of the sub-pixel opening on the base substrate and an edge of the orthographic projection of the first light-transmitting opening on the base substrate ranges from 1 μm to 3 μm .

For example, in the display substrate provided by at least one embodiment of the present disclosure, the first electrode layer comprises a main body portion and a connection portion, the connection portion is configured to be electrically connected to the pixel driving circuit, and at least part of the main body portion is exposed by the sub-pixel opening; and in a direction parallel to the surface of the base substrate, a plane shape of the main body portion is at least partially same as a plane shape of the sub-pixel opening.

For example, in the display substrate provided by at least one embodiment of the present disclosure, an orthographic projection of the sub-pixel opening on the base substrate is located within an orthographic projection of the main body portion on the base substrate.

For example, in the display substrate provided by at least one embodiment of the present disclosure, a minimum distance between an edge of the orthographic projection of the sub-pixel opening on the base substrate and an edge of the orthographic projection of the main body portion on the base substrate ranges from 1 μm to 5 μm .

For example, in the display substrate provided by at least one embodiment of the present disclosure, an orthographic projection of a first light-transmitting opening corresponding to the sub-pixel opening on the base substrate is located within the orthographic projection of the main body portion on the base substrate.

For example, in the display substrate provided by at least one embodiment of the present disclosure, the orthographic projection of the main body portion on the base substrate is located within an orthographic projection of a first light-transmitting opening corresponding to the sub-pixel opening on the base substrate.

For example, the display substrate provided by at least one embodiment of the present disclosure further comprises a color film layer, the color film layer comprises a plurality of color film patterns, and the plurality of color film patterns are respectively provided in the plurality of first light-transmitting openings.

For example, in the display substrate provided by at least one embodiment of the present disclosure, the black matrix layer further includes a plurality of second light-transmitting openings, and the plurality of second light-transmitting openings are respectively between the plurality of first

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light-transmitting openings, and the driving circuit layer comprises a plurality of light-transmitting portions; and at least part of the plurality of second light-transmitting openings are in one-to-one correspondence with at least part of the plurality of light-transmitting portions, and are configured to transmit light in a predetermined angle range with the surface of the base substrate.

For example, in the display substrate provided by at least one embodiment of the present disclosure, for a second light-transmitting opening and a light-transmitting portion that are provided corresponding to each other, and in a direction parallel to the surface of the base substrate, a plane size of the second light-transmitting opening is smaller than a plane size of the light-transmitting portion.

For example, in the display substrate provided by at least one embodiment of the present disclosure, for the second light-transmitting opening and the light-transmitting portion that are provided corresponding to each other, an orthographic projection of the second light-transmitting opening on the base substrate at least partially overlaps with an orthographic projection of the light-transmitting portion on the base substrate.

For example, in the display substrate provided by at least one embodiment of the present disclosure, the plurality of sub-pixels comprise a red sub-pixel, a green sub-pixel, and a blue sub-pixel; a first light-transmitting opening exposing a light-emitting device of the red sub-pixel is substantially in a shape of a first ellipse; a first light-transmitting opening exposing a light-emitting device of the green sub-pixel is substantially in a shape of a second ellipse, a length of a long axis of the second ellipse is smaller than a length of a long axis of the first ellipse, and a length of a short axis of the second ellipse is smaller than a length of a short axis of the first ellipse; or, the first light-transmitting opening exposing the light-emitting device of the green sub-pixel is substantially in a shape of a semi-ellipse; and a first light-transmitting opening exposing a light-emitting device of the blue sub-pixel is substantially in a shape of a third ellipse, a length of a long axis of the third ellipse is smaller than the length of the long axis of the first ellipse, and a length of a short axis of the third ellipse is greater than the length of the short axis of the first ellipse.

For example, in the display substrate provided by at least one embodiment of the present disclosure, the first light-transmitting opening exposing the light-emitting device of the red sub-pixel comprises a first arc-shaped edge and a second arc-shaped edge opposite to each other, and a first tip and a second tip at intersections of the first arc-shaped edge and the second arc-shaped edge, and the first tip and the second tip are opposite to each other; the first light-transmitting opening exposing the light-emitting device of the blue sub-pixel comprises a third arc-shaped edge and a fourth arc-shaped edge opposite to each other, and a third tip and a fourth tip at intersections of the third arc-shaped edge and the fourth arc-shaped edge, and the third tip and the fourth tip are opposite to each other; and the first light-transmitting opening exposing the light-emitting device of the green sub-pixel comprises a fifth arc-shaped edge and a fifth tip at an end of the fifth arc-shaped edge.

For example, in the display substrate provided by at least one embodiment of the present disclosure, the sub-pixel opening corresponding to the green sub-pixel comprises a sixth arc-shaped edge and a sixth tip at an end of the sixth arc-shaped edge, a main body portion of a first electrode layer of a light-emitting device of the green sub-pixel comprises a seventh arc-shaped edge, and the seventh arc-shaped edge does not comprise a tip.

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For example, in the display substrate provided by at least one embodiment of the present disclosure, the plurality of sub-pixels comprise a red sub-pixel, a green sub-pixel, and a blue sub-pixel; a first light-transmitting opening exposing a light-emitting device of the red sub-pixel is substantially in a shape of a first racetrack; a first light-transmitting opening exposing a light-emitting device of the green sub-pixel is substantially in a shape of a second racetrack, a length of a long axis of the second racetrack is smaller than a length of a long axis of the first racetrack, and a length of a short axis of the second racetrack is smaller than a length of a short axis of the first racetrack; or, the first light-transmitting opening exposing the light-emitting device of the green sub-pixel is substantially in a shape of a semi-racetrack; and a first light-transmitting opening exposing a light-emitting device of the blue sub-pixel is substantially in a shape of a third racetrack, a length of a long axis of the third racetrack is smaller than the length of the long axis of the first racetrack, and a length of a short axis of the third racetrack is greater than the length of the short axis of the first racetrack.

For example, in the display substrate provided by at least one embodiment of the present disclosure, one red sub-pixel, two green sub-pixels, and one blue sub-pixel form one pixel unit, and a plurality of pixel units formed of the plurality of sub-pixels are arranged in an array on the base substrate.

At least one embodiment of the present disclosure further provides a display device, the display device comprises the display substrate provided by the embodiments of the present disclosure.

For example, display device provided by at least one embodiment of the present disclosure further comprises a texture touch surface and an image sensor array, the image sensor array is on a side of the driving circuit layer away from the light-emitting device layer, and comprises a plurality of image sensors, and the plurality of image sensors are configured to receive light, emitted from a plurality of light-emitting devices in the light-emitting device layer, reflected by a texture on the texture touch surface, and reaching the plurality of image sensors, for texture collection.

BRIEF DESCRIPTION OF THE DRAWINGS

In order to clearly illustrate the technical solution of the embodiments of the present disclosure, the drawings of the embodiments will be briefly described. It is obvious that the described drawings in the following are only related to some embodiments of the present disclosure and thus are not limitative of the present disclosure.

FIG. 1 is a schematic partial cross-sectional view of a display substrate;

FIG. 2 is a schematic plan view of a sub-pixel opening of a pixel definition layer and a sub-pixel light-emitting opening of a black matrix layer of a display substrate;

FIG. 3 is a schematic partial cross-sectional view of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 4A is a schematic plan view of a sub-pixel opening of a pixel definition layer, a first light-transmitting opening of a black matrix layer, and a first electrode layer of a light-emitting device of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 4B is another schematic plan view of a sub-pixel opening of a pixel definition layer, a first light-transmitting opening of a black matrix layer, and a first electrode layer of

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a light-emitting device of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 5 is another schematic partial cross-sectional view of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 6 is a plan layout diagram of a plurality of sub-pixels of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 7 is a plan layout diagram of a plurality of first light-transmitting openings of a black matrix layer corresponding to a plurality of sub-pixels of the display substrate shown in FIG. 6;

FIG. 8A is another plan layout diagram of a plurality of sub-pixels of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 8B is a plan layout diagram of a plurality of first light-transmitting openings of a black matrix layer corresponding to a plurality of sub-pixels of the display substrate shown in FIG. 8A;

FIG. 9 is still another plan layout diagram of a plurality of sub-pixels of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 10 is a plan layout diagram of a plurality of first light-transmitting openings of a black matrix layer corresponding to a plurality of sub-pixels of the display substrate shown in FIG. 9;

FIG. 11A is still another plan layout diagram of a plurality of sub-pixels of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 11B is a plan layout diagram of a plurality of first light-transmitting openings of a black matrix layer corresponding to a plurality of sub-pixels of the display substrate shown in FIG. 11A;

FIG. 12 is a schematic plan view of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 13 is a schematic plan view of a black matrix layer and a color film layer of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 14A is a schematic diagram of a pixel driving circuit of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 14B is a schematic diagram of another pixel driving circuit of a display substrate provided by at least one embodiment of the present disclosure;

FIG. 15-FIG. 21B are schematic partial plan views of respective functional layers of a display substrate and schematic partial plan views of the respective functional layers stacked in sequence provided by at least one embodiment of the present disclosure; and

FIG. 22 is a schematic cross-sectional view of a display device provided by at least one embodiment of the present disclosure.

DETAILED DESCRIPTION

In order to make objects, technical details and advantages of the embodiments of the present disclosure apparent, the technical solutions of the embodiments will be described in a clearly and fully understandable way in connection with the drawings related to the embodiments of the present disclosure. Apparently, the described embodiments are just a part but not all of the embodiments of the present disclosure. Based on the described embodiments herein, those skilled in the art can obtain other embodiment(s), without any inventive work, which should be within the scope of the present disclosure.

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Unless otherwise defined, all the technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art to which the present disclosure belongs. The terms “first,” “second,” and similar terms are not intended to indicate any sequence, amount or importance, but distinguish various components. The terms “comprise,” “comprising,” “include,” “including,” etc., are intended to specify that the elements or the objects stated before these terms encompass the elements or the objects and equivalents thereof listed after these terms, but do not preclude the other elements or objects. The terms “connect,” “connected,” etc., are not intended to define a physical connection or mechanical connection, but may include an electrical connection, directly or indirectly. “On,” “under,” “left,” “right” and the like are only used to indicate relative position relationship, and when the position of the object which is described is changed, the relative position relationship may be changed accordingly.

In order to prevent the screen from reflecting light, in the traditional OLED display substrate, a layer of polarizer is usually attached on the display substrate to improve the use comfort of the display substrate under ambient light. However, the inventor(s) of the present disclosure found that the transmittance of the polarizer is usually only about 40%, resulting in a low light extraction rate of the display substrate, which in turn leads to higher power consumption of the display substrate.

In some embodiments, COE (Cover film On Encapsulation) technology, that is, a technology that uses a color film (CF) to replace the polarizer, can be used to improve the light extraction rate of the display substrate, and this technology is beneficial to the development of the display substrate in the direction of high integration and thinness.

For example, FIG. 1 shows a schematic partial cross-sectional view of an exemplary display substrate using COE technology. As shown in FIG. 1, the display substrate has structures such as a pixel definition layer E, a light-emitting device, a black matrix layer C, and an encapsulation layer F. The pixel definition layer E has a sub-pixel opening E1, the sub-pixel opening E1 exposes an anode D of the light-emitting device, and a light-emitting layer B1 and a cathode B2 of the light-emitting device are formed in the sub-pixel opening E1 and on the anode D. Within the range defined by the sub-pixel opening E1, the light-emitting layer B1 is in contact with the anode D, and the light-emitting layer B1 can be jointly driven by the anode D and the cathode B2 to emit light. Thus, the region defined by the sub-pixel opening E1 is an effective light-emitting region of the sub-pixel. The encapsulation layer F is provided on the light-emitting device, the black matrix layer C is provided on the encapsulation layer F, and the black matrix layer C has a sub-pixel light-exiting opening C1, which is used to expose the effective light-emitting region of the sub-pixel, so that the light emitted by the light-emitting device of the sub-pixel can be emitted out. For example, a color film A is formed in the sub-pixel light-exiting opening C1, and the color of the color film A is the same as the color of the light emitted by the light-emitting layer of the light-emitting device, thereby improving the light-exiting purity of the display substrate and improving the light extraction rate of the display substrate. Alternatively, the light-emitting layer of the light-emitting device emits white light, and after adding the color film A, monochromatic light can be formed.

However, the inventor(s) of the present disclosure found that, as shown in FIG. 1, the external light (as shown by the arrow in FIG. 1) will be diffracted through the edge of the sub-pixel light-exiting opening C1, and part light of the

external light reflected by the anode and cathode of the light-emitting device will also be diffracted through the edge of the sub-pixel light-exiting opening C1, resulting in the color separation phenomenon of the display substrate, that is, the phenomenon in which the external light and the part light of external light reflected by the anode and the cathode of the light-emitting device generate color at the edge of the sub-pixel light-exiting opening C1. Through research, it is found that the diffracted light range (the shape of the diffracted light) generated by the above-mentioned diffraction phenomenon is related to the shape and size of the sub-pixel light-exiting opening C1.

For example, FIG. 2 shows a schematic plan view of a sub-pixel opening of a pixel definition layer and a sub-pixel light-exiting opening of a black matrix layer corresponding to one sub-pixel of an exemplary display substrate. As shown in FIG. 2, the plane shape of the region defined by the sub-pixel opening E1 of the pixel definition layer E is hexagonal, and correspondingly, the plane shape of the region defined by the sub-pixel light-exiting opening C1 of the black matrix layer C is also hexagonal. Because the sub-pixel light-exiting opening C1 of the black matrix layer C in the display substrate using COE technology is small, for example, in the order of $\lambda \cdot 10^2$, in the current sub-pixel arrangement, for example, in the sub-pixel arrangement with red sub-pixels, green sub-pixels and blue sub-pixels, at the hexagonal sub-pixel light-exiting opening C1, the display substrate will inevitably generate the diffraction effect of monochromatic light (red, green and blue, etc.) under external light (such as a point light source). In addition, because the light-emitting efficiencies of the light-emitting devices of different color sub-pixels are different, the shapes and sizes of the sub-pixel openings E1 of the pixel definition layer E corresponding to the different color sub-pixels are usually different, in which the diffraction phenomena generated by sub-pixels with narrower opening size and sub-pixels with shorter opening size are more serious, and these diffraction phenomena further aggravate the degree of color separation phenomenon.

It should be noted that, in the embodiments of the present disclosure, the color separation phenomenon refers to the phenomenon that the reflected light presents a separation of colors (e.g., red, green, and blue) under external light (e.g., point light source, or line light source) when the display substrate is in the off-screen state.

At least one embodiment of the present disclosure provides a display substrate and a display device. The display substrate has a plurality of sub-pixels arranged in an array, and includes a base substrate, a driving circuit layer on the base substrate, a pixel definition layer and a light-emitting device layer on a side of the driving circuit layer away from the base substrate, and a black matrix layer on a side of the light-emitting device layer away from the base substrate. Each of the plurality of sub-pixels includes a pixel driving circuit in the driving circuit layer and a light-emitting device in the light-emitting device layer, and the pixel driving circuit is configured to drive the light-emitting device. The pixel definition layer includes a plurality of sub-pixel openings, the light-emitting device includes a first electrode layer, a light-emitting material layer, and a second electrode layer that are sequentially stacked in a direction away from the base substrate, the pixel definition layer is on a side of the first electrode layer away from the base substrate, and the plurality of sub-pixel openings respectively expose first electrode layers of the light-emitting devices of the plurality of sub-pixels. The black matrix layer includes a plurality of first light-transmitting openings respectively exposing the

light-emitting devices of the plurality of sub-pixels in a direction perpendicular to a surface of the base substrate, and at least one of the plurality of first light-transmitting openings has an arc-shaped edge. In the direction perpendicular to the surface of the base substrate, at least part of the plurality of sub-pixel openings are in one-to-one correspondence with and at least partially overlapped with the plurality of first light-transmitting openings.

In the above-mentioned display substrate provided by at least one embodiment of the present disclosure, at least one of the plurality of first light-transmitting openings has an arc-shaped edge, and the arc-shaped edge can reduce or even eliminate the phenomenon that the external light is diffracted at the edge of the first light-transmitting opening of the black matrix layer to cause color separation of the display substrate, thereby improving the display effect of the display substrate.

Hereinafter, the display substrate and the display device provided by the embodiments of the present disclosure will be described in detail through several specific embodiments.

FIG. 3 shows a schematic cross-sectional view of a display substrate provided by at least one embodiment of the present disclosure. As shown in FIG. 3, the display substrate has a plurality of sub-pixels arranged in an array, and FIG. 3 shows one sub-pixel as an example. The display substrate includes a base substrate 101, a driving circuit layer 102 provided on the base substrate 101, a light-emitting device layer provided on a side of the driving circuit layer 102 away from the base substrate 101, and a black matrix layer 113 provided on a side of the light-emitting device layer away from the base substrate 101.

As shown in FIG. 3, each sub-pixel includes a pixel driving circuit provided in the driving circuit layer 102 and a light-emitting device EM provided in the light-emitting device layer, and the pixel driving circuit is configured to drive the light-emitting device EM. The black matrix layer 113 includes a plurality of first light-transmitting openings 1131 respectively exposing the light-emitting devices EM of the plurality of sub-pixels in a direction (i.e., in the vertical direction in the figure) perpendicular to the surface of the base substrate 101, so as to respectively transmit light emitted by the light-emitting devices EM of the plurality of sub-pixels. For example, FIG. 4A shows a schematic plan view of the first light-transmitting opening 1131, that is, a schematic plan view in a direction parallel to the surface of the base substrate 101. As shown in FIG. 4A, at least one first light-transmitting opening 1131 has an arc-shaped edge, for example, each first light-transmitting opening 1131 has an arc-shaped edge.

For example, in some embodiments, as shown in FIG. 4A, in the direction parallel to the surface of the base substrate 101, the plane shape of at least one (e.g., each) of the plurality of first light-transmitting openings 1131 is substantially elliptical (or mango-shaped), semi-elliptical, circular, semi-circular, racetrack-shaped (the case shown in the figure), semi-racetrack-shaped, or a deformed shape thereof.

It should be noted that, in the embodiments of the present disclosure, the racetrack shape refers to a racetrack-like shape formed by a rectangle and two arcs on opposite sides of the rectangle, and the racetrack shape has two straight sides arranged parallel to each other and two circular arcs arranged opposite to each other. The mango shape may be regarded as a deformed shape of an ellipse with two arc edges arranged opposite to each other. For details, please refer to FIG. 6 and FIG. 7 described later.

For example, as shown in FIG. 3, the pixel driving circuit of each sub-pixel includes at least one thin film transistor

TFT and a storage capacitor Cst. The thin film transistor TFT includes an active layer **1021**, a gate electrode **1022**, a source electrode **1023**, a drain electrode **1024**, and the like. The source electrode **1023** of the thin film transistor TFT is electrically connected to the first electrode layer **104** of the light-emitting device EM. For example, the storage capacitor Cst includes a first capacitor electrode C1 and a second capacitor electrode C2. For example, the first capacitor electrode C1 of the storage capacitor Cst is provided in the same layer as the gate electrode **1022** of the thin film transistor TFT.

For example, the pixel driving circuit is formed into a 2T1C (two thin film transistors and one storage capacitor) structure, a 6T1C (six thin film transistors and one storage capacitor) structure, or the like, and therefore include a plurality of thin film transistors, and the plurality of thin film transistors have a stacked structure similar or identical to that of the thin film transistor shown in FIG. 3. FIG. 3 only shows the thin film transistor directly connected to the light-emitting device, and the thin film transistor may be a driving thin film transistor, a light-emitting control thin film transistor, or the like.

In addition, it should be noted that, in the embodiments of the present disclosure, "provided in the same layer" means that two functional layers or structural layers are formed in the same layer and with the same material in the hierarchical structure of the display substrate, that is, during the manufacturing process, the two functional layers or structural layers are formed by the same material layer, and the required patterns and structures are formed by the same patterning process.

In addition, as shown in FIG. 3, the display panel further includes a buffer layer **103** provided on the base substrate **101**, a first gate insulating layer **1024** provided on the active layer **1021**, a second gate insulating layer **1025** provided on the gate electrode **1022** and the first capacitor electrode C1, an interlayer insulating layer **1026** provided on the second capacitor electrode CE2, a passivation layer **1027** provided on the source electrode **1023** and the drain electrode **1024**, a planarization layer **109** provided on the passivation layer **1027**, and the like.

For example, in some embodiments, as shown in FIG. 3, the display substrate further includes a pixel definition layer **108** provided on the side of the driving circuit layer **102** away from the base substrate **101**, for example, the pixel definition layer **108** is provided on the planarization layer **109**, and the pixel definition layer **108** includes a plurality of sub-pixel openings **1081**. The light-emitting device EM includes a first electrode layer **104**, a light-emitting material layer **105**, and a second electrode layer **106** that are sequentially stacked in a direction away from the base substrate **101**, the pixel definition layer **108** is provided on a side of the first electrode layer **104** away from the base substrate **101**, and the plurality of sub-pixel openings **1081** respectively expose the first electrode layers **104** of the light-emitting devices EM of the plurality of sub-pixels. In the direction perpendicular to the surface of the base substrate **101**, that is, in the vertical direction in the figure, the plurality of sub-pixel openings **1081** are in one-to-one correspondence with and at least partially overlapped with the plurality of first light-transmitting openings **1131**. Thus, the light emitted by the light-emitting device EM can be emitted out from the first light-transmitting opening **1131** to achieve the display effect.

For example, in some embodiments, as shown in FIG. 4A, in the direction parallel to the surface of the base substrate **101**, the plane shape of the at least one sub-pixel opening

1081 is substantially elliptical (or mango-shaped), semi-elliptical, circular, semi-circular, racetrack-shaped (the case shown in the figure), semi-racetrack-shaped, or a deformed shape thereof.

For example, in some embodiments, as shown in FIG. 4A, for a sub-pixel opening **1081** and a first light-transmitting opening **1131** that are provided corresponding to each other, and in the direction parallel to the surface of the base substrate **101**, the plane shape of the sub-pixel opening **1081** is the same as the plane shape of the first light-transmitting opening **1131**, and both are shown in the figure as a racetrack shape.

For example, in some embodiments, as shown in FIG. 4A, the orthographic projection of the sub-pixel opening **1081** on the base substrate **101** is located within the orthographic projection of the first light-transmitting opening **1131** on the base substrate **101**; that is, the plane size of the sub-pixel opening **1081** is smaller than the plane size of the first light-transmitting opening **1131**.

Because the light-emitting material layer **105** is in contact with the first electrode layer **104** within the range defined by the sub-pixel opening **1081**, the light-emitting material layer **105** can be jointly driven by the first electrode layer **104** and the second electrode layer **106** to emit light. Thus, the region defined by the sub-pixel opening **1081** is the effective light-emitting region of the sub-pixel. By designing the plane shape of the sub-pixel opening **1081** to be substantially the same as the plane shape of the first light-transmitting opening **1131**, and the plane size of the sub-pixel opening **1081** is smaller than the plane size of the first light-transmitting opening **1131**, the effective light-emitting region of the sub-pixel is sufficiently exposed by the first light-transmitting opening **1131**, and the light emitted by the light-emitting device of the sub-pixel can be sufficiently emitted out from the first light-transmitting opening **1131**. Thus, the display substrate can fully utilize the light emitted by the light-emitting devices of the sub-pixels for display, thereby improving the light extraction rate of the display substrate and saving power consumption.

For example, in some embodiments, as shown in FIG. 4A, the minimum distance D1 between an edge of the orthographic projection of the sub-pixel opening **1081** on the base substrate **101** and an edge of the orthographic projection of the first light-transmitting opening **1131** on the base substrate **101** ranges from 1 μm to 3 μm , such as 1.5 μm , 2 μm , or 2.5 μm , etc. That is, the sub-pixel opening **1081** is shrunk inward by 1 μm -3 μm relative to the first light-transmitting opening **1131**, so that the effective light-emitting region defined by the sub-pixel opening **1081** is sufficiently exposed by the first light-transmitting opening **1131**.

For example, in some embodiments, as shown in FIG. 3 and FIG. 4A, the first electrode layer **104** includes a main body portion **1041** and a connection portion **1042**, the connection portion **1042** is configured to be electrically connected to the pixel driving circuit, and at least part of the main body portion **1041** is exposed by the sub-pixel opening **1081**. For example, in the direction parallel to the surface of the base substrate **101**, the plane shape of the main body portion **1041** is the same as the plane shape of the sub-pixel opening **1081**.

For example, in some embodiments, as shown in FIG. 4A, the orthographic projection of the sub-pixel opening **1081** on the base substrate **101** is located within the orthographic projection of the main body portion **1041** on the base substrate **101**. Thus, the shape and size of the main body portion **1041** exposed by the sub-pixel opening **1081** are the same as the shape and size of the sub-pixel opening **1081**.

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Therefore, during the manufacturing process, the effective light-emitting area of each sub-pixel can be obtained by designing the size of the sub-pixel opening **1081**. In addition, the larger size of the main body portion **1041** also provides a margin for the possible positional deviation of the sub-pixel opening **1081** during the manufacturing process.

For example, in some embodiments, as shown in FIG. 4A, the minimum distance D2 between an edge of the orthographic projection of the sub-pixel opening **1081** on the base substrate **101** and an edge of the orthographic projection of the main body portion **1041** on the base substrate **101** ranges from 1 μm to 5 μm , such as 2.5 μm , 3 μm , or 3.5 μm , etc. That is, the sub-pixel opening **1081** is shrunk inward by 1 μm -5 μm relative to the main body portion **1041**.

For example, in some embodiments, as shown in FIG. 4A, the orthographic projection of the first light-transmitting opening **1131** corresponding to the sub-pixel opening **1081** on the base substrate **101** is located within the orthographic projection of the main body portion **1041** on the base substrate **101**, that is, in the direction parallel to the surface of the base substrate **101**, the plane sizes of the main body portion **1041**, the first light-transmitting opening **1131**, and the sub-pixel opening **1081** gradually decrease. The design is beneficial to improve the production yield of the display substrate, improve the light extraction rate of the display substrate, and reduce or even eliminate the phenomenon of color separation of the display substrate.

For example, in other embodiments, as shown in FIG. 4B, the orthographic projection of the main body portion **1041** on the base substrate **101** is located within the orthographic projection of the first light-transmitting opening **1131** corresponding to the sub-pixel opening **1081** on the base substrate **101**. In this case, the shapes of the first light-transmitting opening **1131** and the main body portion **1041** are basically the same, and the first light-transmitting opening **1131** expands outward relative to the main body portion **1041**. This solution can also improve the light extraction rate of the display substrate and reduce or even eliminate the phenomenon of color separation of the display substrate.

For example, in some embodiments, as shown in FIG. 3, the display substrate further includes other structures such as a spacer **107** provided on the pixel definition layer **108** and an encapsulation layer EN provided on the light-emitting device EM of the sub-pixel. For example, the encapsulation layer EN includes a plurality of sub-encapsulation layers to improve its encapsulation effect. For example, the encapsulation layer EN is a composite encapsulation layer, including a first inorganic encapsulation layer **110**, a second organic encapsulation layer **111**, and a third inorganic encapsulation layer **112**. For example, the first inorganic encapsulation layer **110** and the third inorganic encapsulation layer **112** are formed of an inorganic material such as silicon nitride, silicon oxide, silicon oxynitride, or the like, and the second organic encapsulation layer **111** is formed of an organic material such as polyimide (PI), epoxy resin, or the like. The composite encapsulation layer can form multiple protections for the functional structures on the display substrate, and has a better encapsulation effect.

For example, in other embodiments, as shown in FIG. 5, the display substrate further includes a connection electrode **1043**, the first electrode layer **104** of the light-emitting device EM of the sub-pixel is electrically connected to the source electrode **1023** of the thin film transistor TFT through the connection electrode **1043**. For example, another planarization layer **1091** is further formed on the connection electrode **1043**, and in this case, the pixel definition layer **108** is provided on the planarization layer **1091**. For other

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structures of the display substrate shown in FIG. 5, references may be made to the description of the display substrate shown in FIG. 3 and FIG. 4A, which will not be repeated here.

For example, in the embodiments of the present disclosure, the base substrate **101** includes a flexible insulating material such as polyimide (PI) or a rigid insulating material such as a glass substrate. For example, in some examples, the base substrate **101** is a stacked structure in which a plurality of flexible layers and a plurality of barrier layers are alternately arranged. In this case, the flexible layer may include polyimide, and the barrier layer may include an inorganic insulating material such as silicon oxide, silicon nitride, silicon oxynitride, or the like. For example, the buffer layer **103** may include an inorganic material such as silicon nitride, silicon oxide, silicon oxynitride, or the like. The active layer **1021** is made of a material such as polysilicon, metal oxide, or the like, and the first gate insulating layer **1024** and the second gate insulating layer **1025** are made of an inorganic insulating material such as silicon oxide, silicon nitride, silicon oxynitride, or the like. The gate electrode **1022** and the first capacitor electrode **C1** are made of a metal material such as copper, aluminum, titanium, cobalt, or the like, for example, they are formed into a single-layer structure or a multi-layer structure, for example a multi-layer structure such as titanium/aluminum/titanium, molybdenum/aluminum/molybdenum, or the like. The second capacitor electrode **C2** is made of a metal such as copper, aluminum, titanium, cobalt, or the like, or an alloy material; the interlayer insulating layer **1026** is made of an inorganic insulating material such as silicon oxide, silicon nitride, silicon oxynitride, or the like; and the passivation layer **1027** is made of an inorganic insulating material such as silicon oxide, silicon nitride, silicon oxynitride, or the like. For example, in some embodiments, the display substrate may also have no passivation layer as shown in FIG. 3 and FIG. 5. The source and drain electrodes **1023** and **1024** are made of a metal material such as copper, aluminum, titanium, cobalt, or the like, for example, are formed into a single-layer structure or a multi-layer structure, for example a multi-layer structure such as titanium/aluminum/titanium, molybdenum/aluminum/molybdenum, or the like. The first electrode layer **104** is, for example, an anode layer, including a metal oxide such as ITO, IZO, or the like, or a metal such as Ag, Al, Mo, or the like, or their alloys. The material of the light-emitting material layer **105** is an organic light-emitting material. For example, the material of the light-emitting material layer **105** is selected as a light-emitting material that can emit light of a certain color (e.g., red light, blue light, green light, etc.) according to requirements. The second electrode layer **106** is, for example, a cathode layer, including a metal such as Mg, Ca, Li, Al, or the like, or their alloys, or a metal oxide such as IZO, ZTO, or the like, or an organic material with conductive properties such as PEDOT/PSS (poly3, 4-ethylenedioxythiophene/polystyrene sulfonate), or the like. The planarization layer **109** (and the planarization layer **1091**), the pixel definition layer **108**, and the spacer **107** are made of an organic insulating material such as polyimide or the like. The embodiments of the present disclosure do not specifically limit the material of each functional layer.

For example, in some embodiments, as shown in FIG. 3, the display substrate further includes a color film layer **114**, the color film layer **114** includes a plurality of color film patterns **1141**, and the plurality of color film patterns **1141** are respectively provided in the plurality of first light-transmitting openings **1131**. Therefore, the light emitted by

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the light-emitting device EM of the sub-pixel can be emitted out through the color film pattern **1141** to improve the purity of the emitted light.

For example, as shown in FIG. 3, the display substrate further includes a protective cover plate **115** provided on the black matrix layer **113** and the color film layer **114** to protect the structure of the display substrate. For example, the protective cover plate **115** may be a glass cover plate, which can be bonded to the display substrate by optically transparent glue (not shown in the figure).

For example, in some embodiments, as shown in FIG. 3, the black matrix layer **113** may further include a plurality of second light-transmitting openings **1132**, the plurality of second light-transmitting openings **1132** are respectively arranged between the plurality of first light-transmitting openings **1131**, and the driving circuit layer includes a plurality of light-transmitting portions **1020**; at least part of the second light-transmitting openings **1132** are provided in one-to-one correspondence with at least part of the plurality of light-transmitting portions **1020**, and are configured to transmit light which is in a predetermined angle range with the surface of the base substrate **101**, for example, transmit the light L shown in the figure. Thus, the light L can pass through the display substrate from a display side (the upper side in the figure) to a non-display side (the lower side in the figure) of the display substrate, so as to be used for a photosensitive device (such as an image sensor, etc.) that may be provided on the non-display side of the display substrate for photosensitive operation.

For example, the plurality of light-transmitting portions **1020** include a light-transmitting insulating material, and the light-transmitting insulating material includes the light-transmitting insulating material of an insulating layer such as the first gate insulating layer **1024**, the second gate insulating layer **1025**, the interlayer insulating layer **1026**, the passivation layer **1027**, or the like.

For example, in some embodiments, for a second light-transmitting opening **1132** and a light-transmitting portion **1020** that are provided corresponding to each other, and in the direction parallel to the surface of the base substrate, the plane size of the second light-transmitting opening **1132** is smaller than the plane size of the light-transmitting portion **1020**, which will be described in detail later.

For example, in some embodiments, for the second light-transmitting opening **1132** and the light-transmitting portion **1020** that are provided corresponding to each other, the orthographic projection of the second light-transmitting opening **1132** on the base substrate **101** at least partially overlaps with the orthographic projection of the light-transmitting portion **1020** on the base substrate **101**, which will be described in detail later.

For example, in some embodiments, as shown in FIG. 6, the plurality of sub-pixels of the display substrate include a red sub-pixels R, a green sub-pixel G, and a blue sub-pixel B, the pixel definition layer includes a red sub-pixel opening **11**, a green sub-pixel opening **12**, and a blue sub-pixel opening **13**, and the light-emitting devices EM of the red sub-pixel R, the green sub-pixel G and the blue sub-pixel B are respectively formed in the red sub-pixel opening **11**, the green sub-pixel opening **12** and the blue sub-pixel opening **13** of the pixel definition layer.

For example, FIG. 7 shows the plurality of first light-transmitting openings **1131** of the black matrix layer **113** corresponding to the plurality of sub-pixels in FIG. 6. As shown in FIG. 7, the first light-transmitting opening **1131** exposing the light-emitting device EM of the red sub-pixel R is substantially in a shape of a first ellipse **11-1** (or called

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the mango shape, formed by two symmetrical circular arcs), the first light-transmitting opening **1131** exposing the light-emitting device EM of the green sub-pixel G is substantially in a shape of a second ellipse **12-1** (or called the mango shape), the length L2 of the long axis of the second ellipse **12-1** is smaller than the length L1 of the long axis of the first ellipse **11-1**, and the length W2 of the short axis of the second ellipse **12-1** is smaller than the length W1 of the short axis of the first ellipse **11-1**. The first light-transmitting opening **1131** exposing the light-emitting device of the blue sub-pixel B is substantially in a shape of a third ellipse **13-1**, the length L3 of the long axis of the third ellipse **13-1** is smaller than the length L1 of the long axis of the first ellipse **11-1**, and the length W3 of the short axis of the third ellipse **13-1** is greater than the length L1 of the short axis of the first ellipse **11-1**.

For example, as shown in FIG. 6 and FIG. 7, the shape of the first light-transmitting opening **1131** exposing the light-emitting device EM of the red sub-pixel R is the same as the shape of the red sub-pixel opening **11** of the pixel definition layer corresponding to the red sub-pixel R, and the plane size of the red sub-pixel opening **11** is smaller than the plane size of the first light-transmitting opening **1131** exposing the light-emitting device EM of the red sub-pixel R. The shape of the first light-transmitting opening **1131** exposing the light-emitting device EM of the green sub-pixel G is the same as the shape of the green sub-pixel opening **12** of the pixel definition layer corresponding to the green sub-pixel G, and the plane size of the green sub-pixel opening **12** is smaller than the plane size of the first light-transmitting opening **1131** exposing the light-emitting device EM of the green sub-pixel G. The shape of the first light-transmitting opening **1131** exposing the light-emitting device EM of the blue sub-pixel B is the same as the shape of the blue sub-pixel opening **13** of the pixel definition layer corresponding to the blue sub-pixel B, and the plane size of the blue sub-pixel opening **13** is smaller than the plane size of the first light-transmitting opening **1131** exposing the light-emitting device EM of the blue sub-pixel B.

By performing Lab color space detection on the above-mentioned display substrate including the sub-pixel with the shape of an ellipse (or mango shape), for example, using a color analyzer to detect the display substrate in a dark state, it is concluded that the Lab value is 7.68, while the Lab value of the display substrate including the hexagonal sub-pixel (as shown in FIG. 2) is 28.3. The lower the Lab value, the less the phenomenon of color separation on the display substrate. Therefore, the display substrate provided by the embodiments of the present disclosure significantly reduces the degree of color separation of the display substrate.

For example, in other examples, as shown in FIG. 8A and FIG. 8B, the green sub-pixel opening **12** of the pixel definition layer corresponding to the green sub-pixel G is substantially in a shape of a semi-ellipse, the first light-transmitting opening **1311** exposing the light-emitting device of the green sub-pixel G is also substantially in a shape of a semi-ellipse **12-2**, that is, half of an ellipse. For example, the length L21 of the semi-ellipse **12-2** is smaller than the length L1 of the long axis of the first ellipse **11-1**, and the width W21 of the semi-ellipse **12-2** is smaller than or equal to the length W1 of the short axis of the first ellipse **11-1**. For example, the first light-transmitting openings **1311** and the sub-pixel openings of the other sub-pixels in this example are the same as those in FIG. 6 and FIG. 7, and will not be repeated here.

For example, in some examples, as shown in FIG. 6-FIG. 8B, one red sub-pixel R, two green sub-pixels G, and one

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blue sub-pixel B form one pixel unit, and a plurality of pixel units formed of the plurality of sub-pixels are arranged in an array on the base substrate **101**. For example, in other embodiments, one red sub-pixel R, one green sub-pixel G, and one blue sub-pixel B may also form one pixel unit, and a plurality of pixel units formed of the plurality of sub-pixels are arranged in an array on the base substrate **101**, the embodiments of the present disclosure do not limit the specific form of the pixel units.

For example, in other embodiments, as shown in FIG. **9** and FIG. **10**, the plurality of sub-pixels include a red sub-pixel R, a green sub-pixel G, and a blue sub-pixel B. The first light-transmitting opening **1131** exposing the light-emitting device of the red sub-pixel R is substantially in a shape of a first racetrack **21-1**, and the first light-transmitting opening **1131** exposing the light-emitting device of the green sub-pixel G is substantially in a shape of a semi-racetrack **22-1**. For example, the length **L5** of the semi-racetrack **22-1** is smaller than the length **L4** of the long axis of the first racetrack, and the width **W5** of the semi-racetrack **22-1** is greater than or equal to the length **W4** of the short axis of the first racetrack. The first light-transmitting opening **1131** exposing the light-emitting device of the blue sub-pixel B is substantially in a shape of a third racetrack **23-1**, the length **L6** of the long axis of the third racetrack **23-1** is smaller than the length **L4** of the long axis of the first racetrack **21-1**, and the length **W6** of the short axis of the third racetrack **23-1** is greater than the length **W4** of the short axis of the first racetrack **21-1**.

By performing Lab color space detection on the above-mentioned display substrate including the sub-pixel with the shape of a racetrack (or semi-racetrack), it is concluded that the Lab value is 5.18, which is much lower than the lab value, that is 28.3, of the display substrate including the hexagonal sub-pixel (as shown in FIG. **2**).

For example, in other embodiments, as shown in FIG. **11A** and FIG. **11B**, the first light-transmitting opening **1131** exposing the light-emitting device of the green sub-pixel G is substantially in a shape of a second racetrack **22-2**, the length **L7** of the long axis of the second racetrack **22-2** is smaller than the length **L4** of the long axis of the first racetrack **21-1**, and the length **W7** of the short axis of the second racetrack **22-2** is smaller than or equal to the length **W4** of the short axis of the first racetrack **21-1**. Correspondingly, the green sub-pixel opening **22** of the pixel definition layer corresponding to the green sub-pixel G is also in the shape of the second racetrack. The first light-transmitting openings **1311** and the sub-pixel openings of other sub-pixels in this example are the same as those in FIG. **9** and FIG. **10**, and will not be repeated here.

For example, in some examples, as shown in FIG. **9-FIG. 11B**, one red sub-pixel R, two green sub-pixels G, and one blue sub-pixel B form one pixel unit, and a plurality of pixel units formed of the plurality of sub-pixels are arranged in an array on the base substrate **101**. For example, in other embodiments, one red sub-pixel R, one green sub-pixel G, and one blue sub-pixel B may also form one pixel unit, and a plurality of pixel units formed of the plurality of sub-pixels are arranged in an array on the base substrate **101**, the embodiments of the present disclosure do not limit the specific form of the pixel units.

For example, FIG. **12** shows a schematic plan view of a display substrate provided by at least one embodiment of the present disclosure. As shown in FIG. **12**, in this example, one red sub-pixel R, two green sub-pixels G, and one blue sub-pixel B form one pixel unit, and the shapes of the first light-transmitting opening **1131** of the black matrix layer,

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the sub-pixel opening **1081** of the pixel definition layer, and the main body portion **1041** of the first electrode layer **104** corresponding to each sub-pixel are all elliptical (or called mango shape).

For example, as shown in FIG. **7**, the first light-transmitting opening **1131** exposing the light-emitting device of the red sub-pixel includes a first arc-shaped edge **RL1** and a second arc-shaped edge **RL2** opposite to each other, and a first tip **RO1** and a second tip **RO2** located at intersections of the first arc-shaped edge **RL1** and the second arc-shaped edge **RL2**, and the first tip **RO1** and the second tip **RO2** are opposite to each other to form a mango shape.

For example, the first light-transmitting opening **1131** exposing the light-emitting device of the blue sub-pixel includes a third arc-shaped edge **BL1** and a fourth arc-shaped edge **BL2** opposite to each other, and a third tip **BO1** and a fourth tip **BO2** located at intersections of the third arc-shaped edge **BL1** and the fourth arc-shaped edge **BL2**, and the third tip **BO1** and the fourth tip **BO2** are opposite to each other.

For example, the first light-transmitting opening **1131** exposing the light-emitting device of the green sub-pixel includes a fifth arc-shaped edge **GL1** and a fifth tip **GO1** located at an end of the fifth arc-shaped edge **GL1**.

In this case, both the first light-transmitting opening exposing the light-emitting device of the red sub-pixel and the first light-transmitting opening exposing the light-emitting device of the blue sub-pixel have two tips provided opposite to each other, and the first light-transmitting opening exposing the light-emitting device of the green sub-pixel has only one tip; this arrangement can reduce the degree of color separation of the display substrate.

It should be noted that, in the embodiments of the present disclosure, because of the actual process accuracy and process error, the formed tip may not have a sharp corner shape, but the curvature of the tip changes relative to the curvature of the arc-shaped edge, for example, the curvature produces a sudden change at the tip.

For example, as shown in FIG. **6**, the green sub-pixel opening **12** of the pixel definition layer corresponding to the green sub-pixel G includes a sixth arc-shaped edge **GL3** and a sixth tip **GO3** located at an end of the sixth arc-shaped edge **GL3**. As shown in FIG. **12**, the main body portion of the first electrode layer of the light-emitting device of the green sub-pixel includes a seventh arc-shaped edge **GL4**, and the seventh arc-shaped edge **GL4** does not include a tip. This arrangement can reduce the degree of color separation that occurs at the positions where the green sub-pixels are located on the display substrate.

For example, as shown in FIG. **12**, the black matrix layer includes a plurality of second light-transmitting openings **1132**, the driving circuit layer includes a plurality of light-transmitting portions **1020**, and one second light-transmitting opening **1132** corresponds to one light-transmitting portion **1020**; and for a second light-transmitting opening **1132** and a light-transmitting portion **1020** that are provided corresponding to each other, and in the direction parallel to the surface of the base substrate **101**, the plane size of the second light-transmitting opening **1132** is smaller than the plane size of the light-transmitting portion **1020**. For example, in some examples, for the second light-transmitting opening **1132** and the light-transmitting portion **1020** that are provided corresponding to each other, the orthographic projection of the second light-transmitting opening **1132** on the base substrate **101** at least partially overlaps with the orthographic projection of the light-transmitting portion **1020** on the base substrate **101**. For example, the

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orthographic projection of the second light-transmitting opening **1132** on the base substrate **101** is located inside/ within the orthographic projection of the light-transmitting portion **1020** on the base substrate **101**. Thus, as shown in FIG. **3** and FIG. **5**, the light **L** can reach the non-display side (the lower side in the figure) of the display substrate through the second light-transmitting opening **1132** and the light-transmitting portion **1020** in sequence from the display side (the upper side in the figure) of the display substrate, so as to be used for a photosensitive device (such as an image sensor, etc.) that may be provided on the non-display side of the display substrate for photosensitive operation.

For example, in some embodiments, as shown in FIG. **12**, the driving circuit layer **102** further includes a first signal line **S1** and a second signal line **S2** arranged parallel to each other and arranged periodically, and the first signal line **S1** and the second signal line **S2** are configured to provide different electrical signals to the plurality of sub-pixels **SP**. For example, the orthographic projections of the plurality of second light-transmitting openings **1132** on the base substrate **101** are respectively located between the orthographic projection of one first signal line **S1** on the base substrate **101** and the orthographic projection of one second signal line **S2** closest to the one first signal line **S1** on the base substrate **101**.

For example, in some embodiments, the first signal line **S1** is a light-emitting control signal line **EMT**, and the second signal line is a reset voltage line **VNT**, which will be described in detail later.

It should be noted that, considering the process errors and structural errors in actual production, the formed signal line may not be a straight line, such as having uneven portions, etc. In the embodiments of the present disclosure, the first signal line **S1** and the second signal line **S2** are "parallel to each other" means that the angle formed between the extending directions of the first signal line **S1** and the second signal line **S2** is within a range of 0-15 degrees, but not necessarily parallel in the strict sense.

For example, as shown in FIG. **12**, the driving circuit layer includes a third signal line **S3** and a fourth signal line **S4** arranged parallel to each other and arranged periodically, and the third signal line **S3** and the fourth signal line **S4** respectively intersect the first signal line **S1** and the second signal line **S2**, for example, perpendicular to the first signal line **S1** and the second signal line **S2**. The third signal line **S3** and the fourth signal line **S4** are configured to provide different electrical signals to the plurality of sub-pixels, and the orthographic projections of the plurality of second light-transmitting openings **1032** on the base substrate **101** are respectively located between the orthographic projection of one third signal line **S3** on the base substrate **101** and the orthographic projection of one fourth signal line **S4** adjacent to the one third signal line **S3** on the base substrate **101**.

For example, in some embodiments, the third signal line **S3** is a first power supply line **VDD1**, and the fourth signal line **S4** is a data line **DT**, which will be described in detail later.

For example, as shown in FIG. **12**, the first signal line **S1**, the second signal line **S2**, the third signal line **S3**, and the fourth signal line **S4** define a plurality of first regions **RG**, that is, the regions circled by the dotted frame in the figure, and the orthographic projections of the plurality of second light-transmitting openings **1032** on the base substrate **101** are respectively located within the orthographic projections of the plurality of first regions **RG** on the base substrate **101**.

For example, FIG. **13** shows a schematic partial plan view of the black matrix layer and the color film layer of the

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display substrate, and shows a schematic plan view of a plurality of first light-transmitting openings **1131**, a plurality of second light-transmitting openings **1132**, and a plurality of color film patterns **1141**. As shown in FIG. **13**, in the direction perpendicular to the surface of the base substrate **101**, the plurality of color film patterns **1141** include a first color film pattern **1141A** at least partially overlapping with the light-emitting device of a first sub-pixel (e.g., a red sub-pixel) and a second color film pattern **1141B** at least partially overlapping with the light-emitting device of a second sub-pixel (e.g., a green sub-pixel). In the direction parallel to the surface of the base substrate **101**, the plane shape of the first color film pattern **1141A** is different from the plane shape of the second color film pattern **1141B**, and the area of the first color film pattern **1141A** is larger than the area of the second color film pattern **1141B**.

For example, as shown in FIG. **13**, the plane shape of the first color film pattern **1141A** is substantially rectangular, such as a rectangle with a notch, and the plane shape of the second color film pattern **1141B** is substantially semi-elliptical. For example, the area of the first color film pattern **1141A** and the area of the second color film pattern **1141B** are respectively larger than the areas of the first light-transmitting openings **1131** covered by them, so as to fully realize the effect of light filtering.

For example, in some examples, the ratio of the area of the first color film pattern **1141A** to the area of the second color film pattern **1141B** is in a range of (1~1.5):1, such as 1.2:1, 1.4:1, or the like.

For example, as shown in FIG. **13**, in the direction perpendicular to the surface of the base substrate **101**, the plurality of color film patterns **1141** further include a third color film pattern **1141C** at least partially overlapping with the light-emitting device of a third sub-pixel (e.g., a blue sub-pixel). In the direction parallel to the surface of the base substrate **101**, the plane shape of the third color film pattern **1141C** is different from the plane shape of the first color film pattern **1141A** and the plane shape of the second color film pattern **1141B**, and the area of the third color film pattern **1141C** is larger than the area of the first color film pattern **1141A** and the area of the second color film pattern **1141B**, respectively. For example, the plane shape of the third color film pattern **1141C** is an irregular shape, so as to fully realize the effect of light filtering.

For example, in some embodiments, the ratio of the area of the first color film pattern **1141A**, the area of the second color film pattern **1141B**, and the area of the third color film pattern **1141C** is in a range of (1~1.5):1:(1~1.6), such as 1.2:1:1.1, 1.4:1:1.3, or the like.

For example, as shown in FIG. **13**, in the direction perpendicular to the surface of the base substrate **101**, the plurality of color film patterns **1141** further include a fourth color film pattern **1141D** at least partially overlapping with the light-emitting device of a fourth sub-pixel (e.g., a green sub-pixel). In the direction parallel to the surface of the base substrate **101**, the plane shape of the fourth color film pattern **1141D** is substantially the same as the plane shape of the second color film pattern **1141B**, and the area of the fourth color film pattern **1141D** is substantially equal to the area of the second color film pattern **1141B**.

For example, the plane shape of the fourth color film pattern **1141D** is substantially semi-elliptical, and the area of the fourth color film pattern **1141D** is substantially equal to the area of the second color film pattern **1141B**, for example, the difference between the area of the fourth color film

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pattern **1141D** and the area of the second color film pattern **1141B** is not greater than 10% of the area of the second color film pattern **1141B**.

In the embodiments of the present disclosure, the black matrix layer **113** can absorb the light incident into the display substrate, reduce the reflectivity of the display substrate to external light, and improve the display effect of the display substrate; and by covering the black matrix layer **113** with the color film layer **114**, the color film layer **114** can absorb the light incident into the display substrate again to further reduce the reflectivity of the display substrate to external light and improve the display effect of the display substrate. After testing the plurality of color film patterns **1141** shown in FIG. 13, it is found that in the case where the plurality of color film patterns **1141** have the shape and size distribution shown in FIG. 13, the plurality of color film patterns **1141** can fully realize the effect of light filtering as well as the effect of light reflecting, which makes the display effect of the display substrate better.

For example, in some embodiments, as shown in FIG. 13, in the direction perpendicular to the surface of the base substrate **101**, the fourth color film pattern **1141D** partially overlaps with a fourth light-transmitting sub-opening **1132D**.

For example, in some examples, as shown in FIG. 13, the lateral size **1141A-1** of the first color film pattern **1141A** corresponding to the first sub-pixel **P1** ranges from 27 μm to 33 μm , such as 28 μm , 29 μm , 30 μm , or the like, and the longitudinal size **1141A-2** of the first color film pattern **1141A** corresponding to the first sub-pixel **P1** ranges from 30 μm to 35 μm , such as 32 μm , 33 μm , 34 μm , or the like; the lateral size **1141B-1** of the second color film pattern **1141B** corresponding to the second sub-pixel **P2** ranges from 20 μm to 25 μm , such as 21 μm , 22 μm , 23 μm , or the like, and the longitudinal size **1141B-2** of the second color film pattern **1141B** corresponding to the second sub-pixel **P2** ranges from 23 μm to 28 μm , such as 25 μm , 26 μm , 27 μm , or the like; the lateral size **1141C-1** of the third color film pattern **1141C** corresponding to the third sub-pixel **P3** ranges from 32 μm to 38 μm , such as 34 μm , 35 μm , 36 μm , or the like, and the longitudinal size **1141C-2** of the third color film pattern **1141C** corresponding to the third sub-pixel **P3** ranges from 35 μm to 45 μm , such as 38 μm , 40 μm , 42 μm , or the like; and the lateral size **1141D-1** of the fourth color film pattern **1141D** corresponding to the fourth sub-pixel **P4** ranges from 20 μm to 25 μm , such as 21 μm , 22 μm , 23 μm , or the like, and the longitudinal size **1141D-2** of the fourth color film pattern **1141D** corresponding to the fourth sub-pixel **P4** ranges from 23 μm to 28 μm , such as 25 μm , 26 μm , 27 μm , or the like.

For example, in some embodiments, the minimum distance between edges of the plurality of color film patterns **1141** and edges of the plurality of second light-transmitting openings **1132** ranges from 1 μm to 5 μm . For example, as shown in FIG. 3, for the color film pattern **1141** and the second light-transmitting opening **1132** that are at least partially adjacent to each other, there is a space between the color film pattern **1141** and the second light-transmitting opening **1132**, and the minimum distance between the edge of the color film pattern **1141** and the edge of the second light-transmitting opening **1132** ranges from 1 μm to 5 μm , so as to prevent the color film pattern **1141** from filtering the light passing through the second light-transmitting opening **1132**.

For example, referring to FIG. 13 and FIG. 8A, for one color film pattern **1141** and one sub-pixel opening **1081** corresponding to the same sub-pixel, the plane shape of the

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one color film pattern **1141** is different from the plane shape of the one sub-pixel opening **1081**. For example, at least part of the edges of the plurality of second light-transmitting openings **1132** are parallel to at least part of the edges of the color film patterns **1141** adjacent to the plurality of second light-transmitting openings **1132**. For example, in the part shown by the dotted frame in FIG. 13, a part of the edge of the second light-transmitting opening **1132** is parallel to a part of the edge of the color film pattern **1141** adjacent to the second light-transmitting opening **1132**.

Hereinafter, the structure and circuit arrangement of each functional layer of the display substrate provided by the embodiments of the present disclosure will be described in detail through a specific example. In this example, the sub-pixel uses a 7T1C pixel driving circuit to drive the light-emitting device EM.

For example, FIG. 14A shows a circuit diagram of a 7T1C pixel circuit. As shown in FIG. 14A, the pixel circuit includes a driving circuit **122**, a data writing circuit **126**, a compensation circuit **128**, a storage circuit **127**, a first light-emitting control circuit **123**, a second light-emitting control circuit **124**, and a reset circuit **129**.

For example, the driving circuit **122** includes a control terminal **131**, a first terminal **132**, and a second terminal **133**, and the driving circuit **122** is configured to control a driving current flowing through the light-emitting device EM. The control terminal **131** of the driving circuit **122** is connected to a first node N1, the first terminal **132** of the driving circuit **122** is connected to a second node N2, and the second terminal **133** of the driving circuit **122** is connected to a third node N3.

For example, the data writing circuit **126** includes a control terminal, a first terminal, and a second terminal. The control terminal of the data writing circuit **126** is configured to receive a first scan signal, the first terminal of the data writing circuit **126** is configured to receive a data signal, and the second terminal of the data writing circuit **126** is connected to the first terminal **132** (the second node N2) of the driving circuit **122**, and is configured to write the data signal to the first terminal **132** of the driving circuit **122** in response to the first scan signal Ga1. For example, the first terminal of the data writing circuit **126** is connected to a data line **12** to receive the data signal, and the control terminal of the data writing circuit **126** is connected to a scan line **11** to receive the first scan signal Ga1.

For example, during a data writing stage, the data writing circuit **126** is turned on in response to the first scan signal Ga1, so that the data signal is written to the first terminal **132** (the second node N2) of the driving circuit **122**, and the data signal is stored in the storage circuit **127**, and the driving current for driving the light-emitting device EM to emit light is generated according to the data signal during, for example, a light-emitting stage.

For example, the compensation circuit **128** includes a control terminal, a first terminal, and a second terminal. The control terminal of the compensation circuit **128** is configured to receive a second scan signal Ga2, the first terminal and the second terminal of the compensation circuit **128** are electrically connected to the control terminal **131** and the second terminal **133** of the driving circuit **122**, respectively, and the compensation circuit **128** is configured to perform threshold compensation on the driving circuit **120** in response to the second scan signal.

For example, the storage circuit **127** is electrically connected to the control terminal **131** of the driving circuit **122** and a first voltage terminal VDD, and is configured to store the data signal written by the data writing circuit **126**. For

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example, during a data writing and compensation stage, the compensation circuit 128 is turned on in response to the second scan signal Ga2, so that the data signal written by the data writing circuit 126 is stored in the storage circuit 127. For example, still during the data writing and compensation stage, the compensation circuit 128 electrically connects the control terminal 131 and the second terminal 133 of the driving circuit 122, so that the relevant information of the threshold voltage of the driving circuit 122 is also correspondingly stored in the storage circuit, and the stored data signal and the threshold voltage are used to control the driving circuit 122 during, for example, the light-emitting stage, so that the output of the driving circuit 122 is compensated.

For example, the first light-emitting control circuit 123 is connected to the first terminal 132 (the second node N2) of the driving circuit 122 and the first voltage terminal VDD, and is configured to apply a first power supply voltage of the first voltage terminal VDD to the first terminal 132 of the driving circuit 122 in response to a first light-emitting control signal. For example, as shown in FIG. 14A, the first light-emitting control circuit 123 is connected to a first light-emitting control terminal EM1, the first voltage terminal VDD, and the second node N2.

For example, the second light-emitting control circuit 124 is connected to a second light-emitting control terminal EM2, a first terminal 134 of the light-emitting device EM, and the second terminal 132 of the driving circuit 122, and is configured to enable the driving current to be applied to the light-emitting device EM in response to a second light-emitting control signal.

For example, during the light-emitting stage, the second light-emitting control circuit 123 is turned on in response to the second light-emitting control signal provided by the second light-emitting control terminal EM2, so that the driving circuit 122 can apply the driving current to the light-emitting device EM to make it emit light through the second light-emitting control circuit 123; while during the non-light-emitting stage, the second light-emitting control circuit 123 is turned off in response to the second light-emitting control signal, so as to avoid current flowing through the light-emitting device EM to make it emit light, which can improve the contrast of the corresponding display device.

For another example, during an initialization stage, the second light-emitting control circuit 124 is also turned on in response to the second light-emitting control signal, so that the reset circuit is combined to perform a reset operation on the driving circuit 122 and the light-emitting device EM.

For example, the second light-emitting control signal EM2 is the same as or different from the first light-emitting control signal EM1. For example, the second light-emitting control signal EM2 and the first light-emitting control signal EM1 are connected to the same signal output terminal or different signal output terminals, respectively.

For example, the reset circuit 129 is connected to a reset voltage terminal Vinit and the first terminal 134 (a fourth node N4) of the light-emitting device EM, and is configured to apply a reset voltage to the first terminal 134 of the light-emitting device EM in response to a reset signal. In other examples, as shown in FIG. 14A, the reset signal is also applied to the control terminal 131 of the driving circuit, that is, the first node N1. For example, the reset signal is the second scan signal, and the reset signal may also be other signals synchronized with the second scan signal, which is not limited in the embodiments of the present disclosure. For example, as shown in FIG. 14A, the reset circuit 129 is

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respectively connected to the first terminal 134 of the light-emitting device EM, the reset voltage terminal Vinit, and a reset control terminal Rst (a reset control line). For example, during the initialization stage, the reset circuit 129 is turned on in response to the reset signal, so that the reset voltage can be applied to the first terminal 134 of the light-emitting device EM and the first node N1, and the reset operation is performed on the driving circuit 122, the compensation circuit 128, and the light-emitting device EM to eliminate the influence of the previous light-emitting stage.

For example, the light-emitting device EM includes the first terminal 134 and a second terminal 135, the first terminal 134 of the light-emitting device EM is configured to receive the driving current from the second terminal 133 of the driving circuit 122, and the second terminal 135 of the light-emitting device EM is configured to be connected to a second voltage terminal VSS. For example, in one example, as shown in FIG. 14A, the first terminal 134 of the light-emitting device EM is connected to a third node N3 through the second light-emitting control circuit 124. The embodiments of the present disclosure include, but are not limited to, this case. For example, the light-emitting device EM includes various types of OLEDs, such as top emission, bottom emission, double-sided emission, etc., which can emit red, green, blue, or white light, etc., and the first electrode layer and the second electrode layer of the OLED respectively serve as the first terminal 134 and the second terminal 135 of the light-emitting device. The embodiments of the present disclosure do not limit the specific structure of the light-emitting device.

It should be noted that, in the description of the embodiments of the present disclosure, the first node N1, the second node N2, the third node N3, and the fourth node N4 do not necessarily represent actual components, but represent the junctions of related circuit connections in the circuit diagram.

It should be noted that, in the description of the embodiments of the present disclosure, the symbol Vd can represent both the data signal terminal and the level of the data signal. Similarly, the symbols Ga1 and Ga2 can represent both the first scan signal and the second scan signal, as well as the first scan signal terminal and the second scan signal terminal; the symbol Rst can represent both the reset control terminal and the reset signal; the symbol Vinit can represent both the reset voltage terminal and the reset voltage; the symbol VDD can represent both the first voltage terminal and the first power supply voltage; and the symbol VSS can represent both the second voltage terminal and the second power supply voltage. The following embodiments are the same as the above-mentioned case and will not be repeated here.

FIG. 14B is a circuit diagram of a specific implementation example of the pixel circuit shown in FIG. 14A. As shown in FIG. 14B, the pixel circuit includes first to seventh transistors T1, T2, T3, T4, T5, T6, and T7, and includes a storage capacitor Cst. For example, the first transistor T1 is used as a driving transistor, and the other second to seventh transistors are used as switching transistors.

For example, as shown in FIG. 14B, the driving circuit 122 is implemented as the first transistor T1. A gate electrode of the first transistor T1 serves as the control terminal 131 of the driving circuit 122 and is connected to the first node N1; a first electrode of the first transistor T1 serves as the first terminal 132 of the driving circuit 122 and is connected to the second node N2; and a second electrode of the first

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transistor T1 serves as the second terminal 133 of the driving circuit 122 and is connected to the third node N3.

For example, as shown in FIG. 14B, the data writing circuit 126 is implemented as the second transistor T2. A gate electrode of the second transistor T2 is connected to a first scan line (the first scan signal terminal Ga1) to receive the first scan signal, a first electrode of the second transistor T2 is connected to a data line (the data signal terminal Vd) to receive the data signal, and a second electrode of the second transistor T2 is connected to the first terminal 132 (the second node N2) of the driving circuit 122. For example, the second transistor T2 is a P-type transistor, for example, a thin film transistor in which the active layer is formed by low temperature doped polysilicon.

For example, as shown in FIG. 14B, the compensation circuit 128 is implemented as the third transistor T3. A gate electrode of the third transistor T3 is configured to be connected to a second scan line (the second scan signal terminal Ga2) to receive the second scan signal, a first electrode of the third transistor T3 is connected to the control terminal 131 (the first node N1) of the driving circuit 122, and a second electrode of the third transistor T3 is connected to the second terminal 133 (the third node N3) of the driving circuit 122.

For example, as shown in FIG. 14B, the storage circuit 127 is implemented as the storage capacitor Cst, the storage capacitor Cst includes a first capacitor electrode C1 and a second capacitor electrode C2, the first capacitor electrode C1 is connected to the first voltage terminal VDD, and the second capacitor electrode C2 is connected to the control terminal 131 of the driving circuit 122.

For example, as shown in FIG. 14B, the first light-emitting control circuit 123 is implemented as the fourth transistor T4. A gate electrode of the fourth transistor T4 is connected to a first light-emitting control line (the first light-emitting control terminal EM1) to receive the first light-emitting control signal, a first electrode of the fourth transistor T4 is connected to the first voltage terminal VDD to receive the first power supply voltage, and a second electrode of the fourth transistor T4 is connected to the first terminal 132 (the second node N2) of the driving circuit 122.

For example, the light-emitting device EM is implemented as an organic light-emitting diode (OLED), the first electrode layer (here, the anode) of the light-emitting device EM is connected to the fourth node N4 and is configured to receive the driving current from the second terminal 133 of the driving circuit 122 through the second light-emitting control circuit 124, and the second electrode layer (here, the cathode) of the light-emitting device EM is configured to be connected to the second voltage terminal VSS to receive the second power supply voltage. For example, the second voltage terminal is grounded, that is, VSS is at 0V.

For example, the second light-emitting control circuit 124 is implemented as the fifth transistor T5. A gate electrode of the fifth transistor T5 is connected to a second light-emitting control line (the second light-emitting control terminal EM2) to receive the second light-emitting control signal, a first electrode of the fifth transistor T5 is connected to the second terminal 133 (the third node N3) of the driving circuit 122, and a second electrode of the fifth transistor T5 is connected to the first terminal 134 (the fourth node N4) of the light-emitting device EM.

For example, the reset circuit 129 includes a first reset circuit and a second reset circuit, the first reset circuit is configured to apply a first reset voltage Vini1 to the first node N1 in response to a first reset signal Rst1, and the second reset circuit is configured to apply a second reset voltage

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Vini2 to the fourth node N4 in response to a second reset signal Rst2. For example, as shown in FIG. 14B, the first reset circuit is implemented as the sixth transistor T6, and the second reset circuit is implemented as the seventh transistor T7. A gate electrode of the sixth transistor T6 is configured to be connected to a first reset control terminal Rst1 to receive the first reset signal Rst1, a first electrode of the sixth transistor T6 is connected to a first reset voltage terminal Vinit1 to receive the first reset voltage Vini1, and a second electrode of the sixth transistor T6 is configured to be connected to the first node N1. A gate electrode of the seventh transistor T7 is configured to be connected to a second reset control terminal Rst2 to receive the second reset signal Rst2, a first electrode of the seventh transistor T7 is connected to a second reset voltage terminal Vinit2 to receive the second reset voltage Vini2, and a second electrode of the seventh transistor T7 is configured to be connected to the fourth node N4.

It should be noted that each of the transistors used in the embodiments of the present disclosure may be a thin film transistor, a field effect transistor, or other switching components having the same characteristics. In the embodiments of the present disclosure, the thin film transistor is taken as an example for description. The source electrode and drain electrode of the transistor used here may be structurally symmetrical, so that the source electrode and the drain electrode may be structurally indistinguishable. In the embodiments of the present disclosure, in order to distinguish the two electrodes of the transistor except the gate electrode, one electrode is directly described as the first electrode, and the other electrode is described as the second electrode.

For example, referring to FIG. 12, the first signal line S1 is a light-emitting control line EMT, which is used to transmit the above-mentioned first light-emitting control signal EM1 and the second light-emitting control signal EM2. The second signal line S2 is a reset voltage line VNT, which is used to transmit the above-mentioned first reset voltage Vini1 and the second reset voltage Vini2. For example, a side of the reset voltage line VNT away from the light-emitting control line EMT further has a reset control line RST for transmitting the above-mentioned first reset signal Rst1 and second reset signal Rst2.

The layout design of the above-mentioned pixel driving circuit will be described in detail below.

For example, FIG. 15 shows a schematic diagram of a semiconductor layer of the display substrate, the semiconductor layer is used to form the active layers of the thin film transistors T1 to T7 of the pixel driving circuits of the plurality of sub-pixels. FIG. 12 shows pixel driving circuits of two rows of sub-pixels, and the following takes the pixel driving circuits of four directly adjacent sub-pixels (i.e., a first sub-pixel 100a, a second sub-pixel 100b, a third sub-pixel 100c, and a fourth sub-pixel 100d) as an example. The dotted frame in the figure shows the region where the pixel driving circuit of each sub-pixel is located, and the embodiments of the present disclosure are not limited to this layout.

For example, a first gate insulating layer is further provided on the semiconductor layer, which is not shown in the figure, and reference may be made to the first gate insulating layer 1024 in FIG. 3 or FIG. 5.

For example, FIG. 16A shows a schematic diagram of a first gate metal layer of the display substrate, the first gate metal layer is provided on the first gate insulating layer, and FIG. 16B shows a schematic diagram of a stacked layer, in which the first gate metal layer and the semiconductor layer are stacked, of the display substrate.

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For example, as shown in FIG. 16A and FIG. 16B, the first gate metal layer includes a plurality of light-emitting control lines EMT, a plurality of reset control lines RST, a plurality of scan lines GATE, and the first capacitor electrodes C1 of a plurality of storage capacitors Cst, for example, the portions of the light-emitting control line EMT, the reset control line RST, the scan line GATE, and the first capacitor electrode C1 of the storage capacitor Cst overlapping with the active layers of the thin film transistors T1 to T7 constitute the gate electrodes of the thin film transistors T1 to T7. The plurality of light-emitting control lines EMT, the plurality of reset control lines RST, and the plurality of scan lines GATE are respectively electrically connected to the plurality of rows of sub-pixels in one-to-one correspondence to provide corresponding electrical signals.

For example, a second gate insulating layer is further provided on the first gate metal layer, which is not shown in the figure, and reference may be made to the second gate insulating layer 1025 in FIG. 3 and FIG. 5.

FIG. 17A shows a schematic diagram of a second gate metal layer of the display substrate, the second gate metal layer is provided on the second gate insulating layer, and FIG. 17B shows a schematic diagram of a stacked layer, in which the second gate metal layer, the first gate metal layer, and the semiconductor layer are stacked, of the display substrate.

For example, as shown in FIG. 17A and FIG. 17B, the second gate metal layer includes the second capacitor electrode C2 of the storage capacitor Cst and a plurality of reset voltage lines VNT. The second capacitor electrode C2 of the storage capacitor Cst at least partially overlaps with the first capacitor electrode C1 to form a capacitor. The plurality of reset voltage lines VNT are electrically connected to the plurality of rows of sub-pixels in one-to-one correspondence to provide corresponding electrical signals.

For example, an interlayer insulating layer is further provided on the second gate metal layer, which is not shown in the figures, and reference may be made to the interlayer insulating layer 1026 in FIG. 3 and FIG. 5.

FIG. 18A shows a schematic diagram of a first source-drain metal layer of the display substrate, the first source-drain metal layer is provided on the interlayer insulating layer, and FIG. 18B shows a schematic diagram of a stacked layer, in which the first source-drain metal layer, the second gate metal layer, the first gate metal layer, and the semiconductor layer are stacked, of the display substrate.

As shown in FIG. 18A and FIG. 18B, the first source-drain metal layer includes a plurality of first power supply lines VDD1. For example, the plurality of first power supply lines VDD1 are respectively electrically connected to the plurality of columns of sub-pixels in one-to-one correspondence to provide the first power supply voltage. For example, the first source-drain metal layer further includes a plurality of data lines DT. The plurality of data lines DT are electrically connected to the plurality of columns of sub-pixels in one-to-one correspondence to provide data signals. For example, the first source-drain metal layer further includes a plurality of connection electrodes CL for connecting the second capacitor electrode C2 to the first electrode of the third transistor T3, or connecting the first electrode of the sixth transistor T6 to the reset voltage line VNT, or connecting the second electrode of the fifth transistor T5 to the first electrode layer of the light-emitting device, etc.

For example, a passivation layer and a planarization layer are further provided on the first source-drain metal layer, which are not shown in the figures, and reference may be

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made to the passivation layer 1027 and the planarization layer 1091 in FIG. 3 and FIG. 5.

FIG. 19A shows a schematic diagram of a second source-drain metal layer of the display substrate, and the second source-drain metal layer is provided on the planarization layer 1091. FIG. 19B shows a schematic diagram of a stacked layer, in which the second source-drain metal layer, the first source-drain metal layer, the second gate metal layer, the first gate metal layer, and the semiconductor layer are stacked, of the display substrate.

As shown in FIG. 19A and FIG. 19B, the second source-drain metal layer includes a second power supply line VDD2, and the second power supply line VDD2 is in a grid shape. For example, the second power supply line VDD2 is electrically connected to the first power supply line VDD1 to help reduce the resistance on the power supply line, thereby reducing the voltage drop of the power supply line, and to help uniformly transmit the first power supply voltage to each sub-pixel of the display substrate. For example, the second source-drain metal layer further includes a connection electrode 1043 for connecting the first electrode layer of the light-emitting device to the first electrode of the first transistor T1. For example, in the direction perpendicular to the surface of the base substrate, the second power supply line VDD2 at least partially overlaps with the main body portion 1042 of the first electrode layer 104.

For example, another planarization layer, that is, a planarization layer 109, is further provided on the second source-drain metal layer, FIG. 20 shows a schematic plan view of the planarization layer, and in conjunction with FIG. 3 and FIG. 5, the planarization layer 109 has a plurality of vias VA. In this case, the first electrode layer 104 is connected to the connection electrode 1043 through the via VA in the planarization layer 109.

For example, the plurality of vias VA of the planarization layer 109 corresponding to a plurality of sub-pixels located in the same row are not in a straight line. For example, as shown in FIG. 20, one first sub-pixel (e.g., a red sub-pixel), one second sub-pixel (e.g., a green sub-pixel), one third sub-pixel (e.g., a blue sub-pixel), and one fourth subpixel (e.g., a green sub-pixel) that are adjacent to each other and in the same row respectively correspond to vias VA1 to VA4, and the vias VA1 to VA4 are not arranged in the same straight line.

By designing the plurality of vias VA of the planarization layer 109 not to be in a straight line, the lines of the pixel driving circuit can avoid a larger light-transmitting area, thereby forming a light-transmitting portion with a sufficient area.

FIG. 21A shows a schematic diagram of a first electrode material layer of the display substrate, and the first electrode material layer is provided on the passivation layer 109. FIG. 21B shows a schematic diagram of a stacked layer, in which the first electrode material layer, the second source-drain metal layer, the first source-drain metal layer, the second gate metal layer, the first gate metal layer, and the semiconductor layer are stacked, of the display substrate.

As shown in FIG. 21A and FIG. 21B, the first electrode material layer includes the first electrode layers of the light-emitting devices EM of the plurality of sub-pixels, and the first electrode layers of the light-emitting devices EM of the plurality of sub-pixels are connected to the connection electrodes 1043 through the plurality of vias VA in the planarization layers 109, respectively. For example, a light-emitting material layer of the light-emitting device EM is provided on the first electrode layer, and a second electrode layer is provided on the light-emitting material layer.

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For example, other functional layers such as an encapsulation layer, a black matrix layer, a protective cover plate 115, and the like are further formed above the light-emitting device EM, which will not be repeated here.

At least one embodiment of the present disclosure further provides a display device, and FIG. 22 shows a schematic cross-sectional view of the display device. As shown in FIG. 22, the display device includes the display substrate provided by the embodiments of the present disclosure, and the display substrate shown in FIG. 3 is shown in FIG. 22 as an example.

For example, as shown in FIG. 22, the display device further includes a texture touch surface S and an image sensor array 30, for example, the surface of the protective cover plate 115 is implemented as the texture touch surface S. The image sensor array is provided on the side of the driving circuit layer 102 away from the light-emitting device layer, and includes a plurality of image sensors 31 (one image sensor is shown as an example in the figure), and the plurality of image sensors 31 are configured to receive light, which is emitted from the plurality of light-emitting devices EM in the light-emitting device layer, reflected by a texture (e.g., a fingerprint, a palm print, etc.) on the texture touch surface S, and reaching the plurality of image sensors 31, for texture collection.

For example, referring to FIG. 12, the black matrix layer includes the plurality of second light-transmitting openings 1132, the driving circuit layer includes the plurality of light-transmitting portions 1020, and one second light-transmitting opening 1132 corresponds to one light-transmitting portion 1020. In this case, the plurality of image sensors 31 are configured to receive light, which is emitted from the plurality of light-emitting devices EM in the light-emitting device layer, reflected by a texture on the texture touch surface S, and reaching the plurality of image sensors 31 through the plurality of second light-transmitting openings 1132 of the black matrix layer 113 and the plurality of light-transmitting portions 1020 of the driving circuit layer, for texture collection. Thus, the plurality of image sensors 31 can sufficiently receive the light reflected by the texture through the plurality of second light-transmitting openings 1132 and the plurality of light-transmitting portions 1020, thereby increasing the speed of texture recognition and the accuracy of texture recognition.

The display device provided by the embodiments of the present disclosure may further have other structures, and reference may be made to the related art for details, which will not be repeated here.

The following statements should be noted:

- (1) The drawings of the present disclosure involve only the structure(s) in connection with the embodiment(s) of the present disclosure, and other structure(s) can be referred to common design(s).
- (2) For clarity, in the drawings used to describe the embodiments of the present disclosure, the thicknesses of layers or regions are enlarged or reduced, that is, the drawings are not drawn to actual scale. It can be understood that when a component such as a layer, film, region or substrate is referred to as being “on” or “under” another component, the component may be “directly” “on” or “under” another component, or one or more intermediate components may be interposed therebetween.
- (3) In case of no conflict, features in one embodiment or in different embodiments can be combined to obtain new embodiments.

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What have been described above are only specific implementations of the present disclosure, the protection scope of the present disclosure is not limited thereto, and the protection scope of the present disclosure should be based on the protection scope of the claims.

What is claimed is:

1. A display substrate, having a plurality of sub-pixels arranged in an array, and comprising a base substrate, a driving circuit layer on the base substrate, a pixel definition layer and a light-emitting device layer on a side of the driving circuit layer away from the base substrate, and a black matrix layer on a side of the light-emitting device layer away from the base substrate,

wherein each of the plurality of sub-pixels comprises a pixel driving circuit in the driving circuit layer and a light-emitting device in the light-emitting device layer, and the pixel driving circuit is configured to drive the light-emitting device;

the pixel definition layer comprises a plurality of sub-pixel openings, the light-emitting device comprises a first electrode layer, a light-emitting material layer, and a second electrode layer that are sequentially stacked in a direction away from the base substrate, the pixel definition layer is on a side of the first electrode layer away from the base substrate, and the plurality of sub-pixel openings respectively expose first electrode layers of light-emitting devices of the plurality of sub-pixels;

the black matrix layer includes a plurality of first light-transmitting openings respectively exposing the light-emitting devices of the plurality of sub-pixels in a direction perpendicular to a surface of the base substrate, and at least one of the plurality of first light-transmitting openings has an arc-shaped edge; and in the direction perpendicular to the surface of the base substrate, at least part of the plurality of sub-pixel openings are in one-to-one correspondence with and at least partially overlapped with the plurality of first light-transmitting openings;

the first electrode layer comprises a main body portion and a connection portion, the connection portion is configured to be electrically connected to the pixel driving circuit, and at least part of the main body portion is exposed by the sub-pixel opening; and in a direction parallel to the surface of the base substrate, a plane shape of the main body portion is at least partially same as a plane shape of the sub-pixel opening.

2. The display substrate according to claim 1, wherein in a direction parallel to the surface of the base substrate, a plane shape of at least one of the plurality of first light-transmitting openings is elliptical, semi-elliptical, circular, semi-circular, racetrack-shaped, or semi-racetrack-shaped.

3. The display substrate according to claim 1, wherein in a direction parallel to the surface of the base substrate, a plane shape of at least one of the plurality of sub-pixel openings is elliptical, semi-elliptical, circular, semi-circular, racetrack-shaped, or semi-racetrack-shaped.

4. The display substrate according to claim 1, wherein for a sub-pixel opening and a first light-transmitting opening that are provided corresponding to each other, and in a direction parallel to the surface of the base substrate, a plane shape of the sub-pixel opening is same as a plane shape of the first light-transmitting opening.

5. The display substrate according to claim 4, wherein an orthographic projection of the sub-pixel opening on the base substrate is located within an orthographic projection of the first light-transmitting opening on the base substrate.

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6. The display substrate according to claim 5, wherein a minimum distance between an edge of the orthographic projection of the sub-pixel opening on the base substrate and an edge of the orthographic projection of the first light-transmitting opening on the base substrate ranges from 1 μm to 3 μm .

7. The display substrate according to claim 1, wherein an orthographic projection of the sub-pixel opening on the base substrate is located within an orthographic projection of the main body portion on the base substrate;

an orthographic projection of a first light-transmitting opening corresponding to the sub-pixel opening on the base substrate is located within the orthographic projection of the main body portion on the base substrate.

8. The display substrate according to claim 7, wherein a minimum distance between an edge of the orthographic projection of the sub-pixel opening on the base substrate and an edge of the orthographic projection of the main body portion on the base substrate ranges from 1 μm to 5 μm .

9. The display substrate according to claim 8, wherein the orthographic projection of the main body portion on the base substrate is located within an orthographic projection of a first light-transmitting opening corresponding to the sub-pixel opening on the base substrate.

10. The display substrate according to claim 1, further comprising a color film layer, wherein the color film layer comprises a plurality of color film patterns, and the plurality of color film patterns are respectively provided in the plurality of first light-transmitting openings.

11. The display substrate according to claim 1, wherein the black matrix layer further includes a plurality of second light-transmitting openings, and the plurality of second light-transmitting openings are respectively between the plurality of first light-transmitting openings, and the driving circuit layer comprises a plurality of light-transmitting portions; and

at least part of the plurality of second light-transmitting openings are in one-to-one correspondence with at least part of the plurality of light-transmitting portions, and are configured to transmit light in a predetermined angle range with the surface of the base substrate.

12. The display substrate according to claim 11, wherein for a second light-transmitting opening and a light-transmitting portion that are provided corresponding to each other, and in a direction parallel to the surface of the base substrate, a plane size of the second light-transmitting opening is smaller than a plane size of the light-transmitting portion.

13. The display substrate according to claim 12, wherein for the second light-transmitting opening and the light-transmitting portion that are provided corresponding to each other, an orthographic projection of the second light-transmitting opening on the base substrate at least partially overlaps with an orthographic projection of the light-transmitting portion on the base substrate.

14. The display substrate according to claim 1, wherein the plurality of sub-pixels comprise a red sub-pixel, a green sub-pixel, and a blue sub-pixel;

a first light-transmitting opening exposing a light-emitting device of the red sub-pixel is substantially in a shape of a first ellipse;

a first light-transmitting opening exposing a light-emitting device of the green sub-pixel is substantially in a shape of a second ellipse, a length of a long axis of the second ellipse is smaller than a length of a long axis of the first ellipse, and a length of a short axis of the second ellipse is smaller than a length of a short axis of the first ellipse; or, the first light-transmitting opening exposing

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the light-emitting device of the green sub-pixel is substantially in a shape of a semi-ellipse; and

a first light-transmitting opening exposing a light-emitting device of the blue sub-pixel is substantially in a shape of a third ellipse, a length of a long axis of the third ellipse is smaller than the length of the long axis of the first ellipse, and a length of a short axis of the third ellipse is greater than the length of the short axis of the first ellipse.

15. The display substrate according to claim 14, wherein the first light-transmitting opening exposing the light-emitting device of the red sub-pixel comprises a first arc-shaped edge and a second arc-shaped edge opposite to each other, and a first tip and a second tip at intersections of the first arc-shaped edge and the second arc-shaped edge, and the first tip and the second tip are opposite to each other;

the first light-transmitting opening exposing the light-emitting device of the blue sub-pixel comprises a third arc-shaped edge and a fourth arc-shaped edge opposite to each other, and a third tip and a fourth tip at intersections of the third arc-shaped edge and the fourth arc-shaped edge, and the third tip and the fourth tip are opposite to each other; and

the first light-transmitting opening exposing the light-emitting device of the green sub-pixel comprises a fifth arc-shaped edge and a fifth tip at an end of the fifth arc-shaped edge.

16. The display substrate according to claim 15, wherein the sub-pixel opening corresponding to the green sub-pixel comprises a sixth arc-shaped edge and a sixth tip at an end of the sixth arc-shaped edge,

a main body portion of a first electrode layer of a light-emitting device of the green sub-pixel comprises a seventh arc-shaped edge, and the seventh arc-shaped edge does not comprise a tip.

17. The display substrate according to claim 1, wherein the plurality of sub-pixels comprise a red sub-pixel, a green sub-pixel, and a blue sub-pixel;

a first light-transmitting opening exposing a light-emitting device of the red sub-pixel is substantially in a shape of a first racetrack;

a first light-transmitting opening exposing a light-emitting device of the green sub-pixel is substantially in a shape of a second racetrack, a length of a long axis of the second racetrack is smaller than a length of a long axis of the first racetrack, and a length of a short axis of the second racetrack is smaller than a length of a short axis of the first racetrack; or, the first light-transmitting opening exposing the light-emitting device of the green sub-pixel is substantially in a shape of a semi-racetrack; and

a first light-transmitting opening exposing a light-emitting device of the blue sub-pixel is substantially in a shape of a third racetrack, a length of a long axis of the third racetrack is smaller than the length of the long axis of the first racetrack, and a length of a short axis of the third racetrack is greater than the length of the short axis of the first racetrack.

18. The display substrate according to claim 2, wherein one red sub-pixel, two green sub-pixels, and one blue sub-pixel form one pixel unit, and a plurality of pixel units formed of the plurality of sub-pixels are arranged in an array on the base substrate.

19. A display device, comprising the display substrate according to claim 1.

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